

Geophysical and Astrophysical Fluids

Hossein Kafiabad

Laura Currie

October 2024

Table of contents

Preface	3
Module overview	3
Lecture notes	3
 I Michaelmas Term: Geophysical Fluids	 5
1 Introduction	6
1.1 Overview	6
1.2 Review of governing equations	6
1.2.1 Equation of state	7
1.2.2 Incompressible fluids	7
1.2.3 Inviscid fluids	8
1.2.4 Initial and boundary conditions	8
1.3 Hydrostatic balance	8
 2 Stratification	 11
2.1 Buoyancy	11
2.2 Boussinesq equations	13
2.2.1 Density and temperature variation in the ocean and atmosphere	13
2.2.2 Boussinesq approximation	15
2.2.3 Buoyancy as a variable	16
2.2.4 Hydrostatic approximation	18
 3 Shallow Water Systems	 20
3.1 Overview	20
3.2 Derivation	20
3.2.1 Momentum equations	20
3.2.2 Mass continuity equation	22
3.3 Vortices on sloping beaches	25
 4 Rotating fluids	 27
4.1 Rotating reference frame	27
4.2 Dynamics in a rotating frame	29
4.2.1 Centrifugal force as a potential	30
4.3 Tangent-plane approximation	31
4.4 Rotationally dominated flow	34
4.4.1 Geostrophic balance	34
4.4.2 Taylor-Proudman theorem	35
4.4.3 Thermal wind balance	36
 5 Geostrophic theory	 37
5.1 Overview	37

5.2	Scaling the equations and Rossby number	37
5.3	Leading order: geostrophic balance	38
5.3.1	Geostrophic Balance and weather forecast	41
5.4	Next order: quasigeostrophy	43
5.5	Shallow water quasigeostrophy	44
6	Waves in the atmosphere and ocean	46
6.1	Basic concepts	46
6.1.1	Dispersion relation	47
6.1.2	Phase speed and phase velocity	47
6.1.3	Group velocity	49
6.2	Poincaré waves in the Shallow Water equations	50
6.3	Inertia-gravity waves in the Boussinesq equations	52
6.4	Rossby waves	54
7	Turbulence in geophysical flows	57
7.1	Introduction	57
7.2	Fourier transform	59
7.3	Theory of 3D isotropic turbulence	59
7.4	2D turbulence	61
7.5	Geostrophic turbulence	65
7.6	Atmospheric energy spectrum	66

Preface

Module overview

Welcome to Geophysical and Astrophysical Fluids IV.

This module builds on Fluid Mechanics III to introduce the concepts required for understanding the flows that are common in geophysical and astrophysical systems such as the Earth and the Sun.

In Michaelmas, our aim is to derive and understand equations that govern many geophysical flows. We build on the concepts explored in Fluid Mechanics III to introduce the different types of dynamics encountered when rotation and/or stratification are present. We will derive mathematical descriptions and explore the rich landscape of waves and instabilities that can occur in these systems making connections between the mathematics and observed phenomena in planetary atmospheres and oceans along the way.

In Epiphany, we will extend the continuum model of liquids and gases from Fluid Mechanics III to encompass plasmas, through the theory of magnetohydrodynamics (MHD). Because plasmas are ionised gases that conduct electricity, there are now two primary variables: the velocity field and the magnetic field. Their coupling leads not only to important physical effects (such as solar eruptions, planetary magnetospheres, or the failure of fusion reactors) but also to elegant mathematical properties, for whose discovery Hannes Alfvén was awarded a Nobel prize in 1970.

Lecture notes

We will generally follow these notes in the lectures, although not word for word. If you do spot any significant discrepancies, then please let us know! You should also be keeping up to date with the associated problem sheets, found on Learn Ultra.

i In these notes, remarks in boxes like this provide extra information.

Example

These boxes highlight “examples”.

Important Equation Set

Boxes like this one are used to highlight important equation sets.

Below is a table summarising the notation used throughout these notes. The list is not exhaustive.

Table 1: Notation

Symbol	Meaning
$\mathbf{u} = (u, v, w)$	3D fluid velocity and components (in Cartesian coordinates)
$\mathbf{v} = (u, v)$	2D fluid velocity and components (in Cartesian coordinates)
$\nabla = (\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z})$	3D gradient (in Cartesian coordinates)
$\nabla_H = (\frac{\partial}{\partial x}, \frac{\partial}{\partial y})$	2D gradient (in Cartesian coordinates)
$\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$	Laplacian (in Cartesian coordinates)
ρ	Fluid density
ν	Kinematic viscosity
p	Pressure
b	Buoyancy
T	Temperature
$\mathbf{f}_b$	Body force
$N(z)$	Buoyancy (Brunt–Väisälä) frequency

These notes have been created with Quarto (<https://quarto.org/docs/books>).

Part I

Michaelmas Term: Geophysical Fluids

1 Introduction

1.1 Overview

Geophysical fluids, namely air in the atmosphere and water in the ocean, strongly influence life as we know it on planet Earth. Their dynamics shape the weather and the ocean circulations that affect our lives on a daily basis. For example, the daily weather forecasts that we check regularly are basically numerical solutions of dynamical equations for geophysical fluids. Furthermore, climate change is another important aspect relevant to geophysical fluid dynamics. The goal of this course is to study geophysical fluids and their motion from a mathematical point of view.

From Fluid Mechanics III, you will be familiar with the Navier-Stokes equations. Atmospheric and oceanic flows are described by similar equations, but there are two distinguishing features of these flows that are not covered in details in generic fluids courses. The first feature is the variation of density in different layers of the atmosphere and ocean, known as *stratification*. We cover this phenomenon and some of its implications in Chapter 2. The other feature is the effect of the Earth's *rotation*. We are observing the atmospheric and oceanic flows from a rotating frame of reference, which has important implications on their dynamics. In fact, the Earth's rotation is one of the main drivers behind the weather systems and ocean circulation. We introduce rotating fluids and some of their features in Chapter 4.

Geophysical flows are also an amazing platform for the application of mathematics. Due to their complexity, we use a variety of mathematical tools to systematically simplify their governing equations. These simplifications are performed for specific scales or regimes of these fluids. In other words, the simplification and the type of mathematical tools that we employ are problem-dependent. In Chapters 3, 5, and [-@chap-waves], we will use different mathematical tools such as non-dimensionalisation and scaling analysis, perturbation techniques, linearisation and stability analysis in order to study problems of geophysical interest. Hopefully this will equip you with a set of applied maths skills that you can use in other contexts and problems. In the rest of this chapter, we review some generic aspects of fluid dynamics that you should be familiar with from Fluid Mechanics III.

1.2 Review of governing equations

Recall from Fluid Mechanics III, the governing equations for a viscous and compressible fluid are the Navier-Stokes equations:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = \frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{u} = 0, \quad (1.1)$$

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = \mathbf{f}_b - \frac{1}{\rho} \nabla p + \frac{\nu}{3} \nabla (\nabla \cdot \mathbf{u}) + \nu \Delta \mathbf{u}, \quad (1.2)$$

$$+ \text{equation of state} \quad (1.3)$$

where ρ is the fluid density, $\mathbf{u} = (u, v, w)$ is the velocity, p the pressure, ν the kinematic viscosity and $\mathbf{f}_b$ represents body forces acting on the fluid (e.g., gravity). The operator

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla$$

is the material derivative. Equation 1.1 is a result of mass conservation and Equation 1.2 follows from momentum conservation. These four equations (remember $\mathbf{u}$ has three components) provide dynamic relations between ρ , $\mathbf{u}$ and p (5 unknowns). Hence, an equation of state is required to close the system.

1.2.1 Equation of state

The equation of state provides a relationship between the pressure and density that is valid even when the fluid is at rest. This relation will often also depend on the temperature, T , i.e., $p = p(\rho, T)$. In this case, T must be prescribed otherwise we would require a further governing equation for T (which is usually provided by conservation of energy but the details are beyond the scope of this course).

A commonly used equation of state is the ideal gas law $p = \rho \mathcal{R} T$ where $\mathcal{R}$ is a gas constant (which takes a different value for different gases).

Example

Air is considered to satisfy the ideal gas law with $\mathcal{R} = 287 \text{ J kg}^{-1} \text{ K}^{-1}$ (for dry air).

1.2.2 Incompressible fluids

For the incompressible flow we assume that the density of fluid parcels remains constant in time, i.e.,

$$\frac{D\rho}{Dt} = \frac{\partial \rho}{\partial t} + (\mathbf{u} \cdot \nabla)\rho = 0. \quad (1.4)$$

In other words, the rate of change of ρ when following a fluid element is zero but ρ is not necessarily constant. In this case, Equation 1.1 reduces to $\nabla \cdot \mathbf{u} = 0$ and the governing equations for incompressible flow are

$$\frac{\partial \rho}{\partial t} + (\mathbf{u} \cdot \nabla)\rho = 0 \quad (1.5)$$

$$\nabla \cdot \mathbf{u} = 0, \quad (1.6)$$

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = \mathbf{f}_b - \frac{1}{\rho} \nabla p + \nu \Delta \mathbf{u}. \quad (1.7)$$

Note, we have discarded the equation of state as it is redundant for incompressible flow.

i In this course we will be largely concerned with stably stratified fluids where density is not constant and instead it changes with height/depth. However, we can remove a base profile of density and pressure and after a systematic simplification we find $\nabla \cdot \mathbf{u} = 0$ for this system. This simplification is known as *Boussinesq approximation*, which we will study in Chapter 2.

1.2.3 Inviscid fluids

In reality, both air and water are viscous (water is more viscous than air). However, viscosity acts at very small scales. For example, for atmospheric motion, the scale at which viscosity becomes very important (known as the Kolmogorov microscale) ranges from 0.1 to 10mm. One may argue that at scales slightly above 10mm the effect of viscosity might still be felt. However, it is very reasonable to say that for the scales larger than kilometers the effect of viscosity on the dynamics is negligible. For the most parts of this course, we consider the large-scale motion in the atmosphere and ocean – scales of tens of thousand to hundred kilometers. Hence, we safely assume the flow to be inviscid, meaning we set $\nu\Delta\mathbf{u} \approx 0$ in equations (1.2) and (1.7).

That being said, viscosity is still an important part of geophysical fluids. Its effect is felt close to the boundaries (near the shores, the bottom of the ocean, or at the ocean surface). Microphysics is another area of geophysical fluids that is highly affected by viscosity (for example in the formation of clouds or disintegration of pollutants) but these are beyond the scope of this course.

1.2.4 Initial and boundary conditions

From the point of view of PDE analysis, the set of Navier Stokes equations (compressible or incompressible) need a set of initial conditions and boundary conditions to have a (unique) solution.

i Even with provided boundary and initial conditions (in the most general case), we are not sure if the Navier-Stokes equations have a unique solution. This problem (the regularity of Navier-Stokes equations) is one of the unsolved millennium maths problems with the prize of one million dollars. There are only seven millennium problems and one of them so far has been solved.

Common boundary conditions which may be used are: 1) *no-slip* condition for viscous fluids in which the velocity is zero or a given value at boundary, and 2) *free-slip* for inviscid flow in which the velocity normal to the boundary is zero. There are other types of boundary conditions such as fixed values of velocity at the flow inlet or zero-shear at free surfaces.

i The governing equations are hard to solve, even numerically! Our approach is to make simplifications. You see different ways of simplifying these equations in different parts of the course. These simplifications in geophysical fluid dynamics are referred to as *reduced models* as they reduce the complexity of original systems.

1.3 Hydrostatic balance

Assume that gravity acts on the fluid so that $\mathbf{f}_b = -g\mathbf{e}_z$ (gravity is acting downwards and z points upwards). For this body force, the vertical momentum equation, which is embedded in Equation 1.7, becomes

$$\frac{\partial w}{\partial t} + (\mathbf{u} \cdot \nabla)w = -\frac{\partial p}{\partial z} - \rho g. \quad (1.8)$$

Note that we also neglected the viscosity effects. In many fluid applications, the vertical velocity is relatively small, motivating the reduction of Equation 1.8 to

$$\frac{\partial p}{\partial z} = -\rho g. \quad (1.9)$$

This equation is called *hydrostatic balance* and is an important concept that will be used throughout this course. From this equation, we see that pressure decreases with height. Specifically, if we fix the horizontal coordinate (x_0, y_0) and integrate from an arbitrary height/depth z upwards/downward to a reference level $z = z_0$ (e.g., ocean surface or ground) then we find

$$\begin{aligned} p(x_0, y_0, z_0) - p(x_0, y_0, z) &= - \int_z^{z_0} \rho(x_0, y_0, s) g \, ds \\ \Rightarrow p(x_0, y_0, z) &= p(x_0, y_0, z_0) + \int_z^{z_0} \rho(x_0, y_0, s) g \, ds. \end{aligned}$$

This says that the pressure at any level is equal to the pressure at the reference level $p_0 = p(x_0, y_0, z_0)$ plus the total weight (per unit area) of the fluid column above that level.

A special case where hydrostatic balance holds perfectly is the fluid at rest ($\mathbf{u} = 0$). In this particular case, Equation 1.6 is trivially satisfied and the horizontal components of Equation 1.7 reduces to

$$0 = -\frac{1}{\rho} \frac{\partial p}{\partial x}, \quad 0 = -\frac{1}{\rho} \frac{\partial p}{\partial y}.$$

Therefore, p is a function of z only. Since the left-hand-side of Equation 1.9 is just a function of z , so must the right-hand-side be, and hence ρ also only depends on z .

Note the equation of state $p = p(\rho, T)$ implies that if p and ρ are only functions of z then so is the temperature T . In cases where T is prescribed, the equation of state can in theory be combined with equation Equation 1.9 to obtain ρ and p as functions of z .

Example

Consider an ideal gas atmosphere with a constant temperature T occupying $z > 0$. By combining the ideal gas law, $\rho = p/\mathcal{R}T$ with the equation for hydrostatic balance (Equation 1.9), we can determine ρ and p (given that T is constant).

From the ideal gas law we have that $\rho = \frac{p}{\mathcal{R}T}$ and therefore, from Equation 1.9, we have

$$\frac{dp}{dz} = -\frac{gp}{\mathcal{R}T} = -\frac{p}{H}, \quad (1.10)$$

where $H = \frac{\mathcal{R}T}{g}$ is a scale height. Solving Equation 1.10 for p gives $p = p_0 e^{-\frac{z}{H}}$ where $p_0 = p(z = 0)$ is the pressure at the bottom of the atmosphere. It follows that $\rho = \frac{p_0}{\mathcal{R}T} e^{-\frac{z}{H}}$.

Therefore, we see that both ρ and p decrease upwards with a scale height of H . A scale height is typically defined as a distance over which a quantity changes by a factor of e .

For the Earth's atmosphere, taking $T = 270\text{K}$ and $\mathcal{R} = 287\text{Jkg}^{-1}\text{K}^{-1}$ gives $H \approx 8\text{km}$.

Example

Consider an ocean of constant temperature T and depth D occupying $-D < z < 0$. In this case, we use an equation of state more appropriate for water:

$$\rho = \bar{\rho}(1 - \alpha_T(T - 273) + \alpha_p p) \quad (1.11)$$

where $\alpha_T > 0$ is the coefficient of thermal expansion and $\alpha_p > 0$ is the coefficient of compression. Estimates for sea-water give $\bar{\rho} = 1028\text{kgm}^{-3}$, $\alpha_T = 1.7 \times 10^{-4}\text{K}^{-1}$ and $\alpha_p = 4.4 \times 10^{-10}\text{Pa}^{-1}$.

Combining this equation of state with the equation for hydrostatic balance (Equation 1.9) gives

$$\frac{1}{\alpha_p \bar{\rho}} \frac{d\rho}{dz} = -\rho g \quad (1.12)$$

$$\Rightarrow \frac{d\rho}{dz} = -\alpha_p \bar{\rho} \rho g = -\frac{\rho}{H}, \quad (1.13)$$

where $H = \frac{1}{g\alpha_p \bar{\rho}}$ is a scale depth. Hence, $\rho = \rho_0 e^{-\frac{z}{H}}$ where $\rho_0 = \rho(z=0)$ is the density at the top of the ocean. So the density increases downwards with a scale depth of H .

For the Earth's ocean, using the values above for α_p and $\bar{\rho}$, we find $H \approx 220\text{km}$. Therefore, over an ocean depth of 5km, the density increases by a factor of $e^{\frac{D}{H}} = e^{\frac{5}{220}} \approx 1.023$, i.e., an increase over only a few percent.

2 Stratification

2.1 Buoyancy

A heavy object sinks down in a fluid because of gravity, but lighter objects float because of an upward force that the fluid exerts on them opposite to the direction of gravity. This force – known as *buoyancy* – is due to the weight of displaced fluid.

Typically, fluids in the atmosphere and ocean are *stratified*. That is, the density of air in the atmosphere and of water in the ocean varies with height. In a stably stratified medium (where lower density layers are above higher density ones), when a fluid parcel is pushed down vertically, we expect that it will try to come back up due to buoyancy force, and vice-versa. By considering the motion of such a fluid parcel, we can infer its buoyancy which is an important property of stratified fluids.

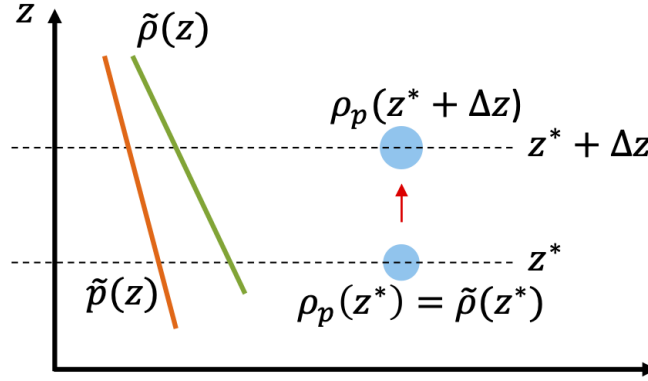


Figure 2.1: A fluid parcel is raised from height z^* to height $z^* + \Delta z$ in a layer of hydrostatically balanced fluid.

Consider a fluid at rest with density $\tilde{\rho}(z)$ and hydrostatic pressure $\tilde{p}$, i.e., $d\tilde{p}/dz = -g\tilde{\rho}$. Then consider raising a parcel of fluid from a height z^* to $z^* + \Delta z$ (see Figure 2.1). At z^* , the environment and the parcel are in hydrostatic balance and so

$$\left. \frac{d\tilde{p}}{dz} \right|_{z^*} = -g\tilde{\rho}(z^*) = -g\rho_p(z^*),$$

where we denote by a subscript p a variable following the parcel and a tilde denotes an environment variable.

The environment at $z^* + \Delta z$ is also in hydrostatic balance:

$$\left. \frac{d\tilde{p}}{dz} \right|_{z^* + \Delta z} = -g\tilde{\rho}(z^* + \Delta z).$$

However, the fluid parcel at $z^* + \Delta z$ is now being acted on by a force given by

$$-\frac{d\tilde{p}}{dz}\Big|_{z^*+\Delta z} - g\rho_p(z^* + \Delta z)$$

since the pressure of the parcel is the same as the pressure of the environment. Note the pressure gradient acts upwards and gravity downwards. Then, since $-g\rho_p(z^* + \Delta z) \neq -g\tilde{\rho}(z^* + \Delta z) = d\tilde{p}/dz\Big|_{z^*+\Delta z}$, there is a net force acting on the fluid parcel (per unit volume) which can be written as

$$F_b = g\tilde{\rho}(z^* + \Delta z) - g\rho_p(z^* + \Delta z) \quad (2.1)$$

We can then apply Newton's Second Law to the parcel at $z^* + \Delta z$ to give

$$\rho_p(z^* + \Delta z) \frac{d^2 \Delta z}{dt^2} = F_b$$

and so

$$\frac{d^2 \Delta z}{dt^2} = \frac{g [\tilde{\rho}(z^* + \Delta z) - \rho_p(z^* + \Delta z)]}{\rho_p(z^* + \Delta z)}. \quad (2.2)$$

Next, if we assume the density of the parcel is conserved then we have that $\rho_p(z^* + \Delta z) = \rho_p(z^*) = \tilde{\rho}(z^*)$. Then we can write Equation 2.2 as

$$\begin{aligned} \frac{d^2 \Delta z}{dt^2} &= \frac{g [\tilde{\rho}(z^* + \Delta z) - \tilde{\rho}(z^*)]}{\tilde{\rho}(z^*)} \\ &\approx \frac{g}{\tilde{\rho}(z^*)} \frac{d\tilde{\rho}}{dz}\Big|_{z^*} \Delta z \end{aligned} \quad (2.3)$$

where we have used a Taylor series for $\tilde{\rho}(z^* + \Delta z)$ to arrive at the final expression. We see that if $d\tilde{\rho}/dz < 0$ then solutions to Equation 2.3 are oscillatory with frequency $N(z^*)$ where in general

$$N^2(z) = -\frac{g}{\tilde{\rho}} \frac{d\tilde{\rho}}{dz}. \quad (2.4)$$

The quantity $N(z)$ is known as the **buoyancy frequency** or **Brunt–Väisälä** frequency. It gives the theoretical frequency of vertical oscillations of a small parcel of fluid in a stratified medium.

Example

Consider a fluid parcel that is raised a distance Δz_0 in a fluid that is at rest. Find the motion of this parcel after it is let go.

We assume that the displacement is small such that $N = N_0$ remains constant. To find the parcel dynamics we then need to solve

$$\frac{d^2 \Delta z}{dt^2} = -N_0^2 \Delta z, \quad \Delta z(t=0) = \Delta z_0, \quad \frac{d\Delta z}{dt}(t=0) = 0. \quad (2.5)$$

The solution to this equation is

$$\Delta z(t) = \Delta z_0 \cos(N_0 t),$$

which is a simple oscillatory motion with frequency N_0 . Equation 2.5 is very similar to the linear pendulum equations. In pendulum motion, the restoring force is gravity, whereas here the restoring force is buoyancy.

i If $d\tilde{\rho}/dz > 0$ then $N^2 < 0$ and the stratification is *unstable* meaning that if we raise a fluid parcel it will continue to rise.

Example

For a constant temperature atmosphere, we saw in an earlier example that $\tilde{\rho} = \rho_s e^{-\frac{z}{H}}$ where $\rho_s = \frac{\rho_0}{RT}$ and $H = \frac{RT}{g}$. For this stratification,

$$N^2 = -\frac{g}{\rho_s e^{-\frac{z}{H}}} \left(-\frac{\rho_s}{H} e^{-\frac{z}{H}} \right) = \frac{g}{H} = \frac{g^2}{RT}.$$

That is, N is independent of z . With $T = 300K$, $N \approx 0.03s^{-1}$. Observations tell us that a more realistic value for N is approximately $0.01s^{-1}$.

2.2 Boussinesq equations

2.2.1 Density and temperature variation in the ocean and atmosphere

In the lower atmosphere and most parts of the ocean, the density (or temperature) does not change significantly at different horizontal locations $\mathbf{x} = (x, y)$ or in time. This motivates the **Boussinesq approximation** which involves writing the density (and pressure) as a sum of a base profile plus small perturbation, where the base profile only depends on z . We will formulate this approximation mathematically in the next section, but before that it will be insightful to look at few observations of density and temperature variations in the ocean and atmosphere.

Figure 2.2 shows a combination of two types of measurements in a region in the arctic ocean. In panel (a), there are four red dots that are **mooring** measurements obtained by a collection of devices connected to a wire and anchored on the sea floor. The other dots are measurements taken by a cruise that sends diving devices down into the ocean. The details of these observations are not important for us, but they show that several locations are sampled by different measuring devices and at different times. Panel (b) shows the density profiles of the cruise measurements at different locations. It is clear that the density profile has a (horizontal) mean that is a function of z . The perturbations to this mean by changing the horizontal location are much smaller than the mean value itself. The variation of density from the base profile is better quantified in the panel (c). The solid black curve shows the mean (base) profile of the density next to a single cast with dashed black curve, which marginally deviates from the mean. Panel (d) shows the density profiles for a fixed location (location D in panel (a)), but at different times. It can be inferred from this plot that changing time does not affect the mean density profile very much. Looking at different part of the ocean, we get a more general base profile (a smoother mean). Nevertheless, the underlying message of this example still holds: the density is well approximated with a base profile that is a function of depth only plus small perturbations.

Figure 2.3 displays the (daily) temperature profiles over a region in India. Atmospheric temperature is directly related to the density at each altitude (which has a fixed pressure). Hence, the relative magnitude of the mean temperature to perturbations can translate to the density as well. It is evident from this figure that changes in temperature at each altitude are relatively small for lower atmosphere (below 15km). There is a jump in temperature that is known as “tropopause”. Near the tropopause, the deviations from the mean temperature profile are larger, and the Boussinesq approximation is not justified.

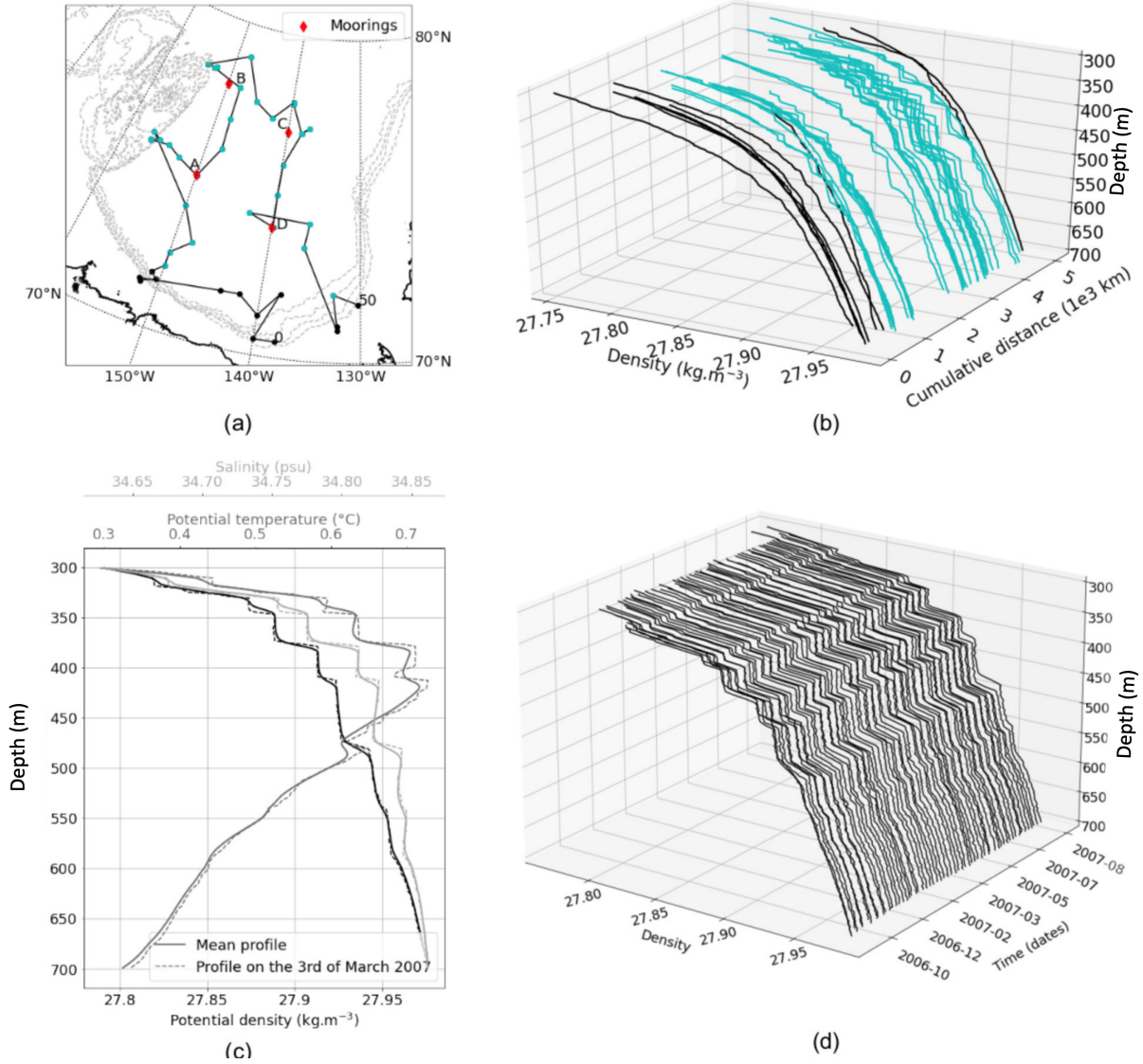


Figure 2.2: (a) Position of different measurements in the Canada Basin of the Arctic Ocean. The trajectory followed during the cruise is represented by the black lines between the cast positions. (b) Density profiles from the 2006 survey within the 300–700m depth range against distance traveled from the first cast. Profiles in blue correspond to the blue dots in (a). (c) Mean temperature and density profiles from mooring D averaged between September 2006 and August 2007 are shown with solid lines. The dashed lines show the same quantities but for a single cast in March 2007. (d) Time evolution of the density profiles at mooring D shown in panel (a). This figure is taken from (Menesguen et al. 2022).

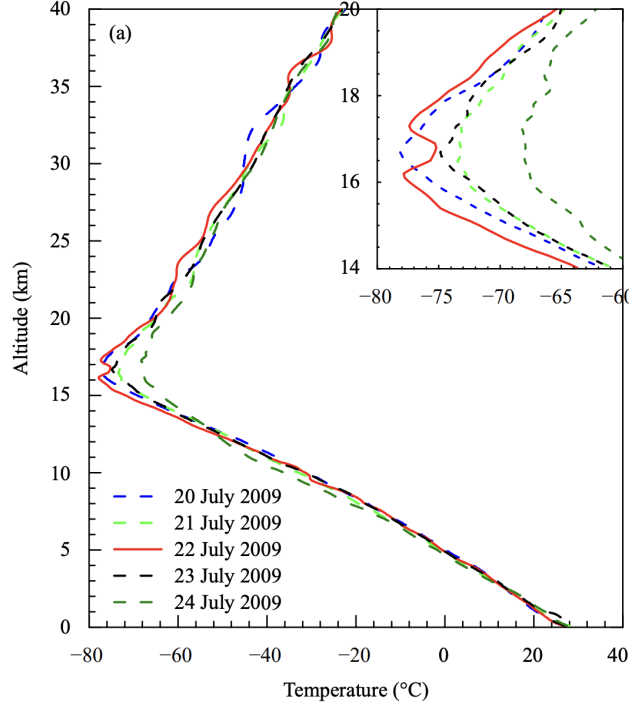


Figure 2.3: An example of temperature-altitude profiles of the daily averaged temperature up to 40 km from 20 to 24 July 2009 over the Indian region (65°N–110°E). The figure is taken from (Prasad et al., 2018).

Measurements similar to the examples just discussed are abundant in the literature thanks to several decades of observational campaigns. Considering them all together, we can conclude that in most parts of the ocean, and in the lower atmosphere, the variations in density at a fixed z are small relative to their means.

2.2.2 Boussinesq approximation

As we have seen, the density in the ocean and in the lower layer of atmosphere varies by only a few percent across the depth or altitude. We can therefore exploit the smallness of these variations and write

$$\rho(\mathbf{x}, t) = \rho_0 + \delta\rho(\mathbf{x}, t)$$

where ρ_0 is the average density (a constant) and $\delta\rho$ are (small) perturbations away from this mean. The perturbations are assumed small so that $|\delta\rho| \ll \rho_0$. Substituting this into the inviscid, incompressible momentum equation (i.e., Equation 1.7 with $\nu = 0$) and again assuming gravity is the only body force acting so that $\mathbf{f}_b = -g\mathbf{e}_z$, we find

$$(\rho_0 + \delta\rho(\mathbf{x}, t)) \frac{D\mathbf{u}}{Dt} = -\nabla p - (\rho_0 + \delta\rho(\mathbf{x}, t))g\mathbf{e}_z.$$

Since $|\delta\rho| \ll \rho_0$ we may be tempted to ignore all the terms involving $\delta\rho$. Instead, we neglect all the terms involving $\delta\rho$ *except* in the term involving gravity (here there is only one but in more complex cases there may be others). This is because while $|\delta\rho| \ll \rho_0$, gravity is strong and so the buoyancy term (the one involving gravity) is retained. Making this approximation, known as the **Boussinesq approximation**, leads to

$$\rho_0 \frac{D\mathbf{u}}{Dt} = -\nabla p - (\rho_0 + \delta\rho(\mathbf{x}, t))g\mathbf{e}_z.$$

Then, dividing by ρ_0 and combining with equation (1.4) and (1.6), we obtain the so called **Boussinesq equations**:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho_0} \nabla p - \frac{\rho}{\rho_0} g \mathbf{e}_z. \quad (2.6)$$

$$\nabla \cdot \mathbf{u} = 0, \quad (2.7)$$

$$\frac{D\rho}{Dt} = \frac{\partial \rho}{\partial t} + (\mathbf{u} \cdot \nabla) \rho = 0, \quad (2.8)$$

i Although we have derived this set of Boussinesq equations starting from the assumption of an incompressible fluid, it is possible to show that the Boussinesq approximation applies to a more general compressible fluid.

i The Boussinesq approximation is valid everywhere in the ocean since density variations are small. However, it is also valid for any motions in the atmosphere that are confined to a layer that is thin compared with the scale height of the atmosphere. For the Earth's atmosphere, we have seen that the scale height is approximately 8km and so the Boussinesq approximation should apply to motions that are confined to layers of about 1km or less.

2.2.3 Buoyancy as a variable

Next, let's consider splitting our variables into a reference state and perturbations to this state. We will consider a static, time-independent reference state with a vertical density stratification $\rho = \tilde{\rho}(z)$. Equation 2.7 and Equation 2.8 are automatically satisfied by this reference state and Equation 2.6 demands

$$p = \tilde{p}(z) \quad \text{with} \quad \frac{d\tilde{p}}{dz} = -\tilde{\rho}g. \quad (2.9)$$

Decomposing our variables in this way gives

$$\mathbf{u} = \mathbf{u}'(\mathbf{x}, t), \quad \rho = \tilde{\rho}(z) + \rho'(\mathbf{x}, t), \quad p = \tilde{p}(z) + p'(\mathbf{x}, t) \quad (2.10)$$

where a tilde denotes a reference state variable and the primed quantities are perturbations to these states (for now no assumptions are made about the size of the perturbations).

Now, substituting (2.10) into Equation 2.6 gives

$$\frac{\partial \mathbf{u}'}{\partial t} + (\mathbf{u}' \cdot \nabla) \mathbf{u}' = -\frac{1}{\rho_0} \nabla p' - \frac{\rho'}{\rho_0} g \mathbf{e}_z = -\frac{1}{\rho_0} \nabla p' + b \mathbf{e}_z,$$

where we have used Equation 2.9 to cancel reference state terms and introduced the *buoyancy* variable

$$b = -\frac{\rho'}{\rho_0} g. \quad (2.11)$$

Note that this expression is analogous to the buoyancy force per unit mass F_b/ρ_0 introduced in Equation 2.1.

Substituting (2.10) into Equation 2.7 simply gives $\nabla \cdot \mathbf{u}' = 0$. Finally, substituting (2.10) into Equation 2.8 gives

$$\frac{\partial(\tilde{\rho}(z) + \rho')}{\partial t} + (\mathbf{u}' \cdot \nabla)(\tilde{\rho}(z) + \rho') = 0$$

$$\Rightarrow \frac{\partial \rho'}{\partial t} + (\mathbf{u}' \cdot \nabla) \rho' + w' \frac{d\tilde{\rho}}{dz} = 0. \quad (2.12)$$

Multiplying Equation 2.12 by $-\frac{g}{\rho_0}$ allows us to form

$$\begin{aligned} \frac{Db}{Dt} + \left(\frac{-g}{\rho_0} \frac{d\tilde{\rho}}{dz} \right) w' &= 0, \\ \Rightarrow \frac{Db}{Dt} + N^2(z) w' &= 0 \end{aligned}$$

where

$$N^2(z) = -\frac{g}{\rho_0} \frac{d\tilde{\rho}}{dz}.$$

Here $N(z)$ is the buoyancy frequency in the Boussinesq approximation. Note it is subtly different to the definition in Equation 2.4.

Gathering together the Boussinesq equations for the perturbation quantities we have the following equation set:

(Non-hydrostatic) Boussinesq Equations

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho_0} \nabla p + b \mathbf{e}_z, \quad (2.13)$$

$$\nabla \cdot \mathbf{u} = 0, \quad (2.14)$$

$$\frac{\partial b}{\partial t} + (\mathbf{u} \cdot \nabla) b + N^2(z) w = 0, \quad (2.15)$$

where we have dropped the primes from perturbation quantities as is customary.

i Note here that p is a perturbation to the background pressure state. In general, when p appears along side buoyancy in the governing equations then this indicates it is a perturbation pressure. This is in contrast to the Boussinesq equations where p in Equation 2.6 is taken to mean the full pressure (likewise in Equation 2.6, ρ is the full density).

Example

The buoyancy (Brunt–Väisälä) frequency that appears in Equation 2.15 is an important quantity in different regimes of geophysical flows. Figure 2.4 shows the values of N^2 for a region in the atmosphere/ocean as functions of altitude/depth. In the left plot, N^2 over the region of Gadanki in India is shown. There is a noticeable jump at around 18km altitude (similar to what we observed in figure Figure 2.3, marking the “tropopause”, which is the border between two layers of atmosphere (troposphere and stratosphere). Note that the height of tropopause varies to some degrees in different locations (that is why the jump in Figure 2.3 and Figure 2.4 appears at slightly different altitudes). Around the tropopause, constant N or even the Boussinesq approximation are not valid, but in the lower atmosphere (below the tropopause) N can be assumed constant. Horizontal bars show standard deviations obtained while considering 43 different measurements during April 2006 to March 2007. The low variability in the lower atmosphere (relatively small bars) justifies the Boussinesq approximation, whereas the high variability near tropopause shows this approximation is no longer valid around this altitude.

The right plot is obtained from a mooring observation (“NISKINE” campaign). For the deep ocean (lower than 800 meters), it is safe to use constant N . However, in the upper ocean the variation of N

with depth cannot be neglected. A comparison between the two figures shows that N is an order of magnitude smaller in the ocean.

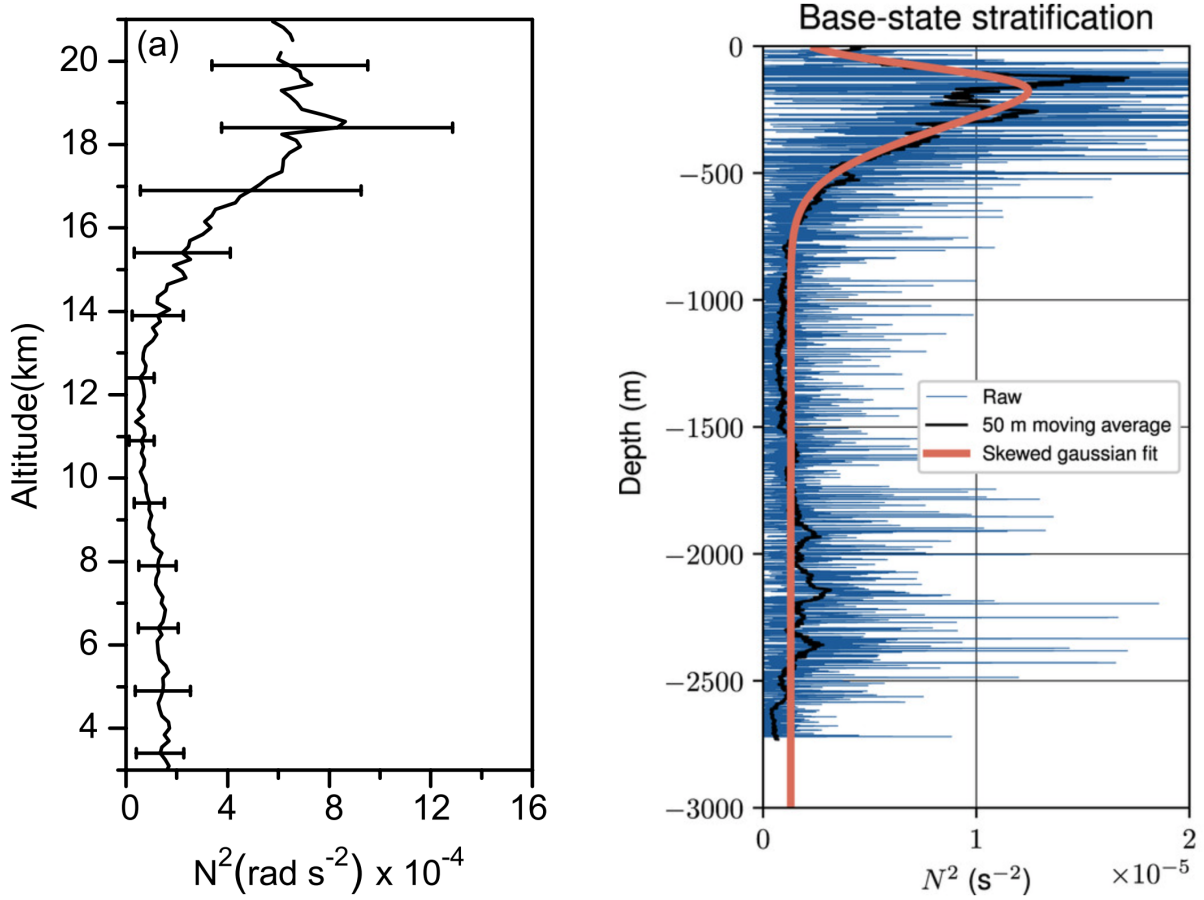


Figure 2.4: Left: Stratification in the atmosphere in the region of Gadanki in India (13.5°N , 79.2°E). This curve is obtained by averaging over 43 radar measurements and the bars show the standard deviation for each altitude. This plot is taken from (Mehta et al., 2008). Right: raw stratification profile (blue) observed during a pilot cruise measurement campaign named “NISKINE”, along with its 50-m moving average (black) and a Gaussian fit (orange). This plot is taken from (Asselin & Young, 2020).

2.2.4 Hydrostatic approximation

A common approximation in geophysical fluid dynamics is to assume that the horizontal scale is large compared to the vertical scale. In this case, the vertical component of the momentum equation is simply replaced by our hydrostatic balance equation. This is known as the *hydrostatic approximation* and leads to the following **hydrostatic Boussinesq equations**

(Hydrostatic) Boussinesq Equations

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{v} = -\frac{1}{\rho_0} \nabla_H p, \quad (2.16)$$

$$\frac{\partial p}{\partial z} = \rho_0 b = -\rho g, \quad (2.17)$$

$$\nabla \cdot \mathbf{u} = 0, \quad (2.18)$$

$$\frac{\partial b}{\partial t} + (\mathbf{u} \cdot \nabla)b + N^2 w = 0, \quad (2.19)$$

where $\mathbf{v} = (u, v)$ is the horizontal velocity and $\nabla_H = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y} \right)$.

Note that the hydrostatic approximation has resulted in the vertical momentum equation being replaced by hydrostatic balance.

i The Boussinesq equations (2.13) - (2.15) (and their hydrostatic counterparts (2.16) - (2.19)) are the fundamental equations that are solved numerically in weather forecast and climate models. Because we observe and measure everything from the rotating Earth, these equations are solved in the rotating frame of reference (which we will come to in Chapter 4). Most climate models use the Hydrostatic equations as they require far fewer computational resources. Hence, because climate models (unlike weather models) are run for very long times (several years or decades), it is worth reducing the computational cost at the expense of making the hydrostatic assumption.

i In the geophysical fluids literature, equations (2.16) - (2.19) are often known as the *primitive equations*. This is somewhat confusing as the non-Hydrostatic Boussinesq equation set is more general. However, the Hydrostatic Boussinesq equations are still more accurate than some other reduced equations such as the shallow water equations and quasigeostrophic equations that we will study in Chapters 3 and 5, respectively.

3 Shallow Water Systems

3.1 Overview

The shallow water equations are a model of a thin layer of fluid with constant density which is bounded by a rigid surface from below and by a free surface from above. By “thinness” – which is the most defining feature of these fluids – we mean the horizontal length scales of these flows are much larger than their vertical scales (and hence the name shallow!). Other characteristics of these equations which we will discuss in this section are the direct consequence of the thinness and constant density assumptions.

The shallow water equations are one of the simplest sets of PDEs describing geophysical fluids, yet they are very useful. They are used in practical models of shallow lakes, rivers and coastal flows. Due to their relative simplicity, many theoretical concepts in geophysical fluid dynamics are first investigated for shallow water systems before being studied in more complex systems. They also incorporate many phenomena that characterise atmospheric and oceanic flows such as geostrophic balance and waves (we will discuss these later in the course). The practical importance of shallow water equations aside, their derivation is a good example of using rigorous mathematical steps and physical assumptions to reduce the Boussinesq equations to a much simpler set.

3.2 Derivation

We use the defining characteristics of shallow water flows to derive their governing equations. Before that, let’s clarify the notation and conventions we are going to use (this is sometimes the most confusing part about shallow water systems because different sources in the literature adopt very different notation). We assume the z axis is perpendicular to the layer of water as shown in Figure 3.1. $z = 0$ can be placed at any arbitrary level (naturally we want to see it close the bottom surface). $\eta(x, y, t)$ is the vertical position of fluid surface measured from $z = 0$. Likewise, $\eta_b(x, y, t)$ is the vertical height of bottom topography measured from $z = 0$. $h(x, y, t)$ is the thickness of water column measured from bottom boundary all the way to free surface, i.e. $h = \eta - \eta_b$. The thinness of fluid has an important consequence: the vertical velocity (in the z direction) is much smaller than the horizontal velocity components (but is not necessarily zero). Hence, we only consider the two-dimensional horizontal velocity denoted by $\mathbf{v} = (u, v)$ and assume $w \ll u, v$. Lastly, we denote the horizontal gradient operator by $\nabla_H = (\partial/\partial x, \partial/\partial y)$.

3.2.1 Momentum equations

One of the main consequences of small vertical velocity appears in the vertical momentum equation. Let’s consider the vertical momentum equation in the Boussinesq system (i.e., Equation 2.6)

$$\frac{\partial w}{\partial t} + (\mathbf{v} \cdot \nabla_H)w + w \frac{\partial w}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial z} - \frac{\rho}{\rho} g. \quad (3.1)$$

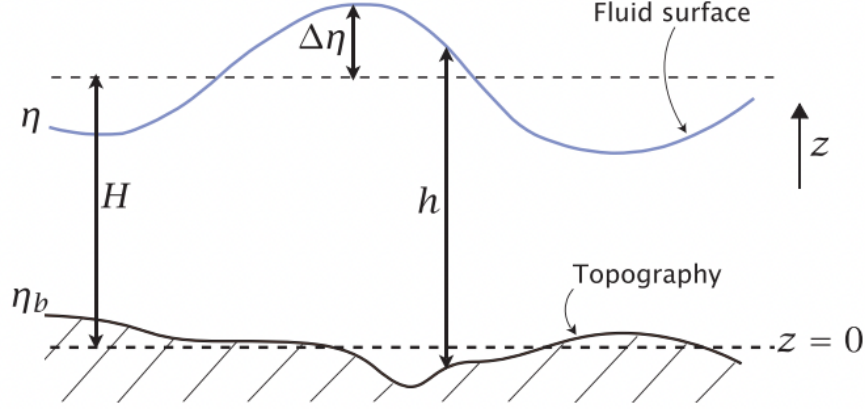


Figure 3.1: A shallow water system. h is the thickness of a water column, H is the mean thickness, η is the height of the free surface and η_b is the height of the lower rigid. $\Delta\eta$ is the deviation of free from average height.

Assuming w to be smaller than the other variables, the vertical momentum equation then reduces to

$$\frac{\partial p}{\partial z} = -\rho g,$$

which is simply the hydrostatic balance that was discussed in Section 1.3 (cf. Equation 1.9). Because the density is assumed constant ($\rho = \rho_0$), we may integrate this equation from η to an arbitrary height z to give

$$p(x, y, z, t) - p(x, y, \eta(x, y, t), t) = \rho_0 g (\eta(x, y, t) - z). \quad (3.2)$$

It is physically justified to assume the pressure at the free surface $p(x, y, \eta, t)$ is constant (for some applications we may need to change this assumption though). Hence, we can take the horizontal gradient of both sides of Equation 3.2 to arrive at

$$\nabla_H p = \rho_0 g \nabla_H \eta. \quad (3.3)$$

Now we consider the horizontal momentum equation in Equation 2.6 and substitute for the horizontal pressure gradient using Equation 3.3

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla_H) \mathbf{v} = -g \nabla_H \eta, \quad (3.4)$$

where we also neglected the vertical velocity at leading order. Note that the right-hand side of this equation is independent of z . Hence, if we assume the velocity is initially independent of z , it should remain independent of z , i.e. $\mathbf{v} = \mathbf{v}(x, y, t)$. As a consequence, for the rest of this chapter we do not deal with z derivative. We also simplify the notation by replacing ∇_H with ∇ in the context of shallow water equation since there is no z dependence. For example, we write Equation 3.5 as

$$\frac{D\mathbf{v}}{Dt} = -g \nabla \eta. \quad (3.5)$$

Note that for three-dimensional flows such as those described by the Boussinesq equations, we still distinguish between horizontal and three-dimensional gradient and show them with ∇_H and ∇ , respectively.

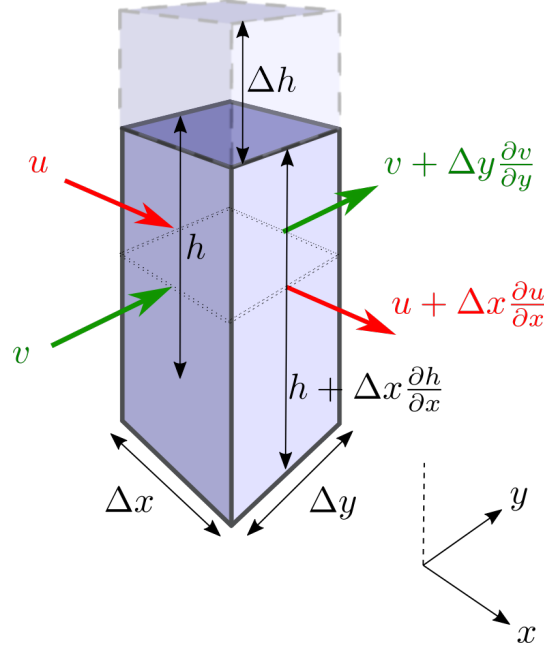


Figure 3.2: A column of fluid in shallow water.

3.2.2 Mass continuity equation

Consider a (Cartesian) column of shallow water as shown in Figure 3.2. The height of this column h is measured from the bottom boundary to the free surface. The cross-sectional area of this column is $\Delta x \Delta y$. If the fluid enters this column from the sides, its height has to increase to conserve mass (note that the density is constant and the fluid is incompressible). More quantitatively, let's consider the two sides perpendicular to the x -axis. If the velocity normal to one of the sides is u , the normal velocity to the other side will be $u + \Delta x \partial u / \partial x$ (see Figure 3.2). Likewise, If the thickness of one side is h , the thickness of the other side will be $h + \Delta x \partial h / \partial x$. The mass flux through a side is calculated as the normal velocity times the surface area of the side times density. For example, if the normal velocity and vertical thickness are u and h respectively, the mass flux will be $\rho_0 u h \Delta y$ (without loss of generality we assume the velocity u is pointing inward). Hence, the total amount of mass entering/leaving from these two sides is

$$\begin{aligned} \text{MF}_x &= \text{mass flux along the } x \text{ axis} = \rho_0 u h \Delta y - \rho_0 \left(u + \Delta x \frac{\partial u}{\partial x} \right) \left(h + \Delta x \frac{\partial h}{\partial x} \right) \Delta y \\ &= -\rho_0 \Delta x \frac{\partial u}{\partial x} h \Delta y - \rho_0 u \Delta x \frac{\partial h}{\partial x} \Delta y - \rho_0 \Delta x \frac{\partial u}{\partial x} \Delta x \frac{\partial h}{\partial x} \Delta y \end{aligned}$$

Likewise, we obtain the total mass flux along the y axis

$$\text{MF}_y = -\rho_0 \Delta y \frac{\partial v}{\partial y} h \Delta x - \rho_0 v \Delta y \frac{\partial h}{\partial y} \Delta x - \rho_0 \Delta y \frac{\partial v}{\partial y} \Delta y \frac{\partial h}{\partial y} \Delta x$$

The rate of change of mass in this fluid column is $\rho_0 \Delta x \Delta y \Delta h / \Delta t$ which should be equal to mass flux from the sides, i.e.,

$$\rho_0 \Delta x \Delta y \frac{\Delta h}{\Delta t} = \text{MF}_x + \text{MF}_y.$$

After dividing this equation by $\rho_0 \Delta x \Delta y$ and taking the limit as $\Delta x, \Delta y, \Delta t \rightarrow 0$, we find

$$\frac{\partial h}{\partial t} = -h \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) - u \frac{\partial h}{\partial x} - v \frac{\partial h}{\partial y}$$

which can be written as

$$\frac{\partial h}{\partial t} + \nabla_{\mathbf{H}} \cdot (h\mathbf{v}) = 0, \quad \text{or} \quad \frac{Dh}{Dt} + h\nabla_{\mathbf{H}} \cdot \mathbf{v} = 0. \quad (3.6)$$

Note that unlike Boussinesq or incompressible flows, the shallow water velocity is not divergence free. This is because the columns of fluids can be compressed by increasing their height.

i The conservation of mass equation for the shallow water system can be derived from 3D mass conservation equation using $w \ll u, v$ (see the assignments).

In summary, in a non-rotating frame, the inviscid shallow water equations are

Inviscid Shallow Water Equations

$$\text{momentum:} \quad \frac{D\mathbf{v}}{Dt} = -g \nabla_{\mathbf{H}} \eta. \quad (3.7)$$

$$\text{mass continuity:} \quad \frac{\partial h}{\partial t} + \nabla_{\mathbf{H}} \cdot (h\mathbf{v}) = 0, \quad \text{or} \quad \frac{Dh}{Dt} + h\nabla_{\mathbf{H}} \cdot \mathbf{v} = 0 \quad (3.8)$$

$$\text{height relation:} \quad h(x, y, t) = \eta(x, y, t) - \eta_b(x, y, t). \quad (3.9)$$

Example

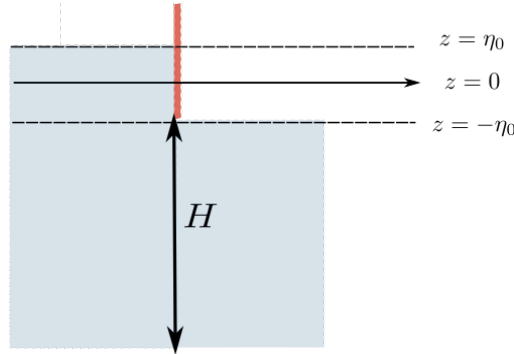


Figure 3.3: A schematic figure of dam break with two levels of water on each side.

Let's consider an example of a dam or membrane that creates two levels of fluid as shown in the figure Figure 3.3. The fluid is initially at rest and the bottom surface is assumed flat (i.e. $\eta_b = \text{constant}$). At $t = 0$, we remove the dam or membrane and want to see how this fluid behaves in time. This flow is one-dimensional so we can simply drop the dependence on y and set $v = 0$. Let's say the difference between two levels is $2\eta_0$ (see the figure). We place $z = 0$ at half distance between the two levels such that

$$\eta(x, t = 0) = \begin{cases} +\eta_0, & x < 0 \\ -\eta_0, & x > 0 \end{cases} = -\eta_0 \operatorname{sgn}(x), \quad (3.10)$$

where sgn is the sign function. Note that the fluid is initially at rest $u(x, t = 0) = 0$. To analyse this problem we assume the disturbances to the initial state are relatively small and write

$$h(x, t) = H + \eta(x, t), \quad u = u(x, t)$$

Substituting these expressions into Equations (3.7)-(3.9) leads to

$$\begin{aligned}\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} &= -g \frac{\partial \eta}{\partial x} \\ \frac{\partial \eta}{\partial t} + \frac{\partial}{\partial x} ((H + \eta)u) &= 0,\end{aligned}$$

and further neglecting the product of (small) disturbance terms (i.e. u and η) gives

$$\frac{\partial u}{\partial t} + g \frac{\partial \eta}{\partial x} = 0 \quad (3.11)$$

$$\frac{\partial \eta}{\partial t} + H \frac{\partial u}{\partial x} = 0. \quad (3.12)$$

We can take the x derivative of Equation 3.11 and t derivative of Equation 3.12 and eliminate u to arrive at

$$\frac{\partial^2 \eta}{\partial t^2} - c^2 \frac{\partial^2 \eta}{\partial x^2} = 0, \quad c = \sqrt{gH},$$

which is simply the one-dimensional wave equation. For unbounded domains, you are familiar with the d'Alembert solution of the wave equation

$$\eta(x, t) = \frac{1}{2} (F(x - ct) + F(x + ct)), \quad (3.13)$$

where $F(x)$ is the initial condition. Using the given initial condition for this problem (Equation 3.10), we find

$$\eta(x, t) = -\frac{1}{2} \eta_0 (\text{sgn}(x - ct) + \text{sgn}(x + ct)) = \begin{cases} +\eta_0 & x < -ct \\ 0 & -ct < x < ct \\ -\eta_0 & x > ct \end{cases}$$

This solution implies that a half of the disturbance moves backwards and a half forwards, both with the speed $c = \sqrt{gH}$. In between the travelling disturbance becomes zero as shown in the left panel of Figure 3.4. It is interesting that the height disturbance travels faster if the water is deeper.

We can also derive the velocity for this solution by using Equation 3.13 and Equation 3.11 (or Equation 3.12)

$$u(x, t) = -\frac{1}{2} \eta_0 \sqrt{\frac{g}{H}} (\text{sgn}(x - ct) - \text{sgn}(x + ct)) = \begin{cases} 0 & x < -ct \\ \eta_0 \sqrt{g/H} & -ct < x < ct \\ 0 & x > ct \end{cases}$$

As the disturbance moves to both sides, it creates a velocity in the middle (see the right panel of Figure 3.4). This velocity is larger in shallower water (proportional to $H^{-1/2}$).

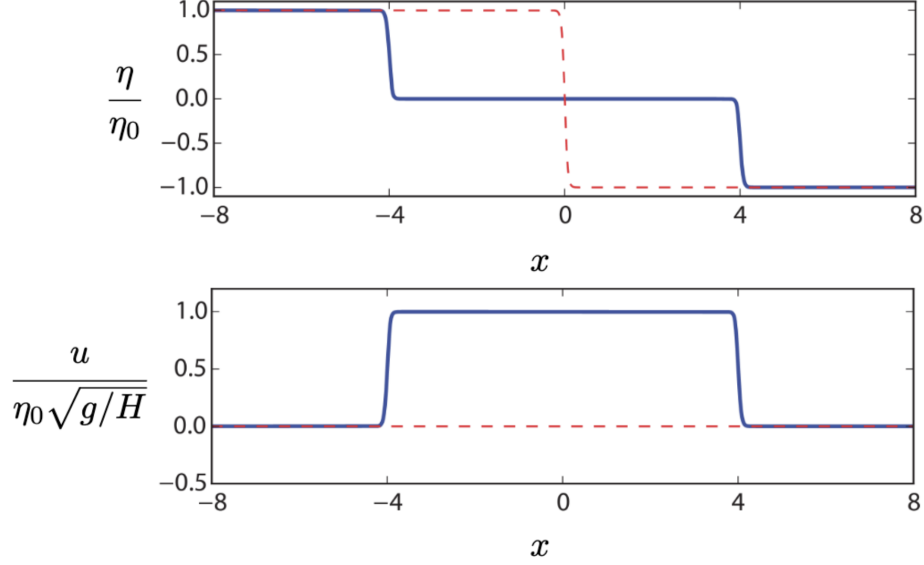


Figure 3.4: The height (top) and velocity (bottom) for the propagating height disturbance in shallow water system. The variables are normalised. The dashed line shows the initial condition and the solid line the solution at $t = 4/c$.

3.3 Vortices on sloping beaches

There are cases where the changes in η are small compared to the flow and bottom surface, but the surface pressure is no longer constant. In such cases, we need to modify the shallow water equations by changing the top boundary condition from constant pressure to constant η . This boundary condition is known as a *rigid lid* as it is literally the equivalent of putting a rigid lid on the top. Using this condition, Equation 3.2 changes to

$$p(x, y, z, t) - p(x, y, 0, t) = \rho_0 g(0 - z),$$

where we place our z axis at the top surface that is nearly constant. Taking the horizontal gradient of this equation gives,

$$\nabla_H p(x, y, z, t) = \nabla_H (p_0(x, y, t) - p(x, y, 0, t)),$$

where we used p_0 to denote the surface pressure. Note the constant η assumption means that $h(x, y, t) = -\eta_b(x, y)$. As a result, if the bottom surface doesn't change with time (reasonable assumption for many problems with a solid boundary), $\partial h / \partial t = 0$. Using these assumptions, we can write the rigid-lid shallow water equations as

$$\frac{D\mathbf{v}}{Dt} = -\frac{1}{\rho_0} \nabla_H p_0, \quad (3.14)$$

$$\nabla_H \cdot (h\mathbf{v}) = 0, \quad (3.15)$$

$$h(x, y, t) = \eta_b(x, y, t). \quad (3.16)$$

i The rigid-lid shallow water system is a good approximation for flows on sloping beaches as surface variations compared to the changes of the bottom slope and velocities are negligible. This is also a good model for lab experiments where the flow is confined by a solid wall from the top.

We can transform equations (3.14)-(3.16) into the following equation (see the assignment for derivation)

$$q = \frac{\nabla \times \mathbf{v}}{h} = \frac{\partial v / \partial x - \partial u / \partial y}{h}, \quad \frac{Dq}{Dt} = 0. \quad (3.17)$$

This shows that the quantity q is conserved materially (i.e. on particle trajectories), which has important consequences for the dynamics. For example, if a vortex is pushed up toward shallower parts of a sloping beach, it becomes stronger and rotates faster.

i q as defined in Equation 3.17 is a special case of a quantity known as Potential Vorticity (PV). We will introduce the general form of this quantity in chapter 5 and show that it is conserved for some limits of the shallow water and Boussinesq equations. Such a conservation law is an important element of large-scale atmospheric and oceanic flows.

4 Rotating fluids

Newton's second law applies in an inertial (non-accelerating) frame of reference but the Earth rotates (i.e., it is not an inertial frame of reference). Considering that we observe and measure everything from Earth's surface, it is useful to describe flows relative to Earth's surface rather than some inertial frame and therefore we need to recast our governing equations into a form appropriate for a rotating frame of reference.

4.1 Rotating reference frame

Consider an inertial frame of reference I, described by orthogonal basis vectors $\mathbf{e}_1$, $\mathbf{e}_2$ and $\mathbf{e}_3$ and a second frame of reference R described by orthogonal basis vectors $\hat{\mathbf{e}}_1$, $\hat{\mathbf{e}}_2$ and $\hat{\mathbf{e}}_3$ which is rotating with angular velocity $\boldsymbol{\Omega}$ relative to I. If we define $\left(\frac{d}{dt}\right)_I$ to be a derivative with respect to time in the inertial frame, then clearly we have that $\left(\frac{d\mathbf{e}_j}{dt}\right)_I = 0$ but $\left(\frac{d\hat{\mathbf{e}}_j}{dt}\right)_I \neq 0$ because $\hat{\mathbf{e}}_j$ is rotating relative to our inertial frame.

Now, consider a vector $\mathbf{C}$ of constant length that is rotating relative to the inertial frame at a constant angular velocity $\boldsymbol{\Omega}$. It follows that, in the rotating frame R, $\mathbf{C}$ appears stationary, i.e., $\left(\frac{d\mathbf{C}}{dt}\right)_R = 0$ where now $\left(\frac{d}{dt}\right)_R$ is a derivative taken in the rotating frame of reference. We want to know how this vector appear to change in time if observed from the inertial frame, i.e., what is $\left(\frac{d\mathbf{C}}{dt}\right)_I$?

Consider a small interval of time δt over which $\mathbf{C}$ rotates through a small angle $\delta\lambda$ (see Figure 4.1). Then, the change in $\mathbf{C}$ as observed in the inertial frame is

$$\delta\mathbf{C} = |\mathbf{C}| \sin \gamma \delta\lambda \mathbf{e}_n$$

where γ is the angle between $\boldsymbol{\Omega}$ and $\mathbf{C}$, and $\hat{n}$ is the unit vector in the direction of the change of $\mathbf{C}$ (perpendicular to both $\boldsymbol{\Omega}$ and $\mathbf{C}$). But the change in λ is easily obtained from the angular velocity, in particular we have $\delta\lambda = |\boldsymbol{\Omega}|\delta t$ and hence

$$\delta\mathbf{C} = |\mathbf{C}||\boldsymbol{\Omega}| \sin \gamma \delta t \mathbf{e}_n = \boldsymbol{\Omega} \times \mathbf{C} \delta t.$$

It follows that

$$\left(\frac{d\mathbf{C}}{dt}\right)_I = \boldsymbol{\Omega} \times \mathbf{C}. \quad (4.1)$$

Example

Consider the case where the rotating frame is rotating about the $\mathbf{e}_3$ axis at a constant rate Ω . In this case basis vectors for the inertial frame and the rotating frame are related by

$$\begin{aligned} \hat{\mathbf{e}}_1 &= \cos(\Omega t)\mathbf{e}_1 + \sin(\Omega t)\mathbf{e}_2, & \hat{\mathbf{e}}_2 &= -\sin(\Omega t)\mathbf{e}_1 + \cos(\Omega t)\mathbf{e}_2, & \hat{\mathbf{e}}_3 &= \mathbf{e}_3, \\ \mathbf{e}_1 &= \cos(\Omega t)\hat{\mathbf{e}}_1 - \sin(\Omega t)\hat{\mathbf{e}}_2, & \mathbf{e}_2 &= \sin(\Omega t)\hat{\mathbf{e}}_1 + \cos(\Omega t)\hat{\mathbf{e}}_2, & \mathbf{e}_3 &= \hat{\mathbf{e}}_3. \end{aligned}$$

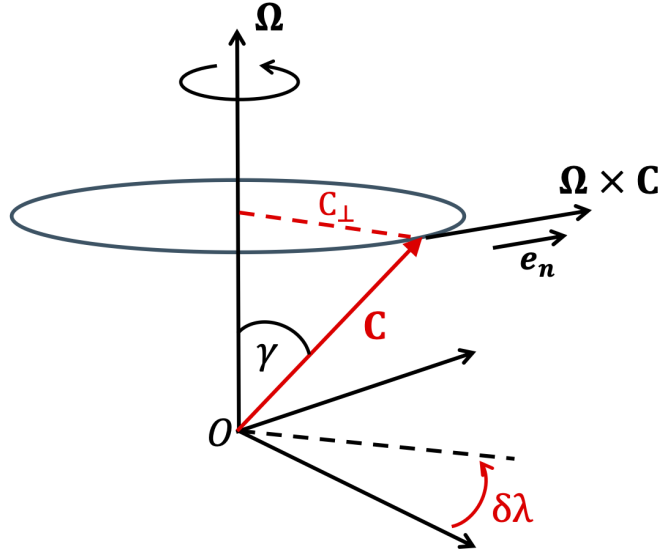


Figure 4.1: A vector $\mathbf{C}$ rotating relative to the inertial frame at a constant angular velocity $\mathbf{\Omega}$. Over a small interval of time δt , $\mathbf{C}$ rotates through a small angle $\delta\lambda$. γ is the angle between $\mathbf{\Omega}$ and $\mathbf{C}$, and $\mathbf{e}_n$ is the unit vector in the direction of the change of $\mathbf{C}$ (perpendicular to both $\mathbf{\Omega}$ and $\mathbf{C}$). Figure adapted from Vallis.

Now, any vector $\mathbf{a}$ can be expressed as

$$\begin{aligned}\mathbf{a} &= a_j \mathbf{e}_j = a_1(\cos(\Omega t)\hat{\mathbf{e}}_1 - \sin(\Omega t)\hat{\mathbf{e}}_2) + a_2(\sin(\Omega t)\hat{\mathbf{e}}_1 + \cos(\Omega t)\hat{\mathbf{e}}_2) + a_3\hat{\mathbf{e}}_3 \\ &= (a_1 \cos(\Omega t) + a_2 \sin(\Omega t))\hat{\mathbf{e}}_1 + (-a_1 \sin(\Omega t) + a_2 \cos(\Omega t))\hat{\mathbf{e}}_2 + a_3\hat{\mathbf{e}}_3 = \hat{a}_j \hat{\mathbf{e}}_j,\end{aligned}$$

where $\hat{a}_1 = a_1 \cos(\Omega t) + a_2 \sin(\Omega t)$, $\hat{a}_2 = -a_1 \sin(\Omega t) + a_2 \cos(\Omega t)$ and $\hat{a}_3 = a_3$. Note the coordinates a_j and $\hat{a}_j$ are different but vector is the same.

We see that even simple trajectories can appear non-trivial in a rotating frame, e.g.,

- A stationary particle with position $\mathbf{a} = (a_0, 0, 0)$ corresponds to $\hat{\mathbf{a}} = (a_0 \cos(\Omega t), -a_0 \sin(\Omega t), 0)$ in the rotating frame. This is a clockwise rotation with angular velocity Ω .
- A particle moving in a straight line at speed v with position $\mathbf{a} = (vt, 0, 0)$ in the inertial frame corresponds to $\hat{\mathbf{a}} = (vt \cos(\Omega t), -vt \sin(\Omega t), 0)$ in the rotating frame. This corresponds to a clockwise spiral.

Note that

$$\begin{aligned}\left(\frac{d\hat{\mathbf{e}}_1}{dt}\right)_I &= \frac{d}{dt}(\cos(\Omega t)\mathbf{e}_1 + \sin(\Omega t)\mathbf{e}_2) = -\Omega \sin(\Omega t)\mathbf{e}_1 + \Omega \cos(\Omega t)\mathbf{e}_2 = \Omega \hat{\mathbf{e}}_2 \\ \left(\frac{d\hat{\mathbf{e}}_2}{dt}\right)_I &= \frac{d}{dt}(-\sin(\Omega t)\mathbf{e}_1 + \cos(\Omega t)\mathbf{e}_2) = -\Omega \cos(\Omega t)\mathbf{e}_1 - \Omega \sin(\Omega t)\mathbf{e}_2 = -\Omega \hat{\mathbf{e}}_1 \\ \left(\frac{d\hat{\mathbf{e}}_3}{dt}\right)_I &= 0.\end{aligned}$$

Combining together, we can write

$$\left(\frac{d\hat{\mathbf{e}}_j}{dt}\right)_I = \mathbf{\Omega} \times \hat{\mathbf{e}}_j$$

which is expected from Equation 4.1.

Now let's consider a general vector $\mathbf{B}(t) = B_j(t)\hat{\mathbf{e}}_j$. In the rotating frame we have that

$$\left(\frac{d\mathbf{B}}{dt}\right)_{\text{R}} = \frac{d}{dt}(B_j\hat{\mathbf{e}}_j)_{\text{R}} = \frac{dB_j}{dt}\hat{\mathbf{e}}_j$$

since the $\hat{\mathbf{e}}_j$ are stationary in the rotating frame. Note, there is no need to state which reference frame we differentiate the B_j in since they are scalar functions.

In the inertial frame we have

$$\begin{aligned}\left(\frac{d\mathbf{B}}{dt}\right)_{\text{I}} &= \frac{d}{dt}(B_j\hat{\mathbf{e}}_j)_{\text{I}} = \frac{dB_j}{dt}\hat{\mathbf{e}}_j + B_j\left(\frac{d\hat{\mathbf{e}}_j}{dt}\right)_{\text{I}} \\ &= \left(\frac{d\mathbf{B}}{dt}\right)_{\text{R}} + B_j\boldsymbol{\Omega} \times \hat{\mathbf{e}}_j \\ &= \left(\frac{d\mathbf{B}}{dt}\right)_{\text{R}} + \boldsymbol{\Omega} \times (B_j\hat{\mathbf{e}}_j) \\ &= \left(\frac{d\mathbf{B}}{dt}\right)_{\text{R}} + \boldsymbol{\Omega} \times \mathbf{B}.\end{aligned}$$

So, for any vector $\mathbf{B}$, we have

$$\left(\frac{d\mathbf{B}}{dt}\right)_{\text{I}} = \left(\frac{d\mathbf{B}}{dt}\right)_{\text{R}} + \boldsymbol{\Omega} \times \mathbf{B}. \quad (4.2)$$

4.2 Dynamics in a rotating frame

If we take $\mathbf{B} = \mathbf{X}$ in Equation 4.2 to be the position vector of a fluid parcel, then we have that

$$\mathbf{u}_{\text{I}} = \mathbf{u}_{\text{R}} + \boldsymbol{\Omega} \times \mathbf{X}$$

where $\mathbf{u}_{\text{I}}$ is the perceived velocity in the inertial frame, and $\mathbf{u}_{\text{R}}$ is the perceived velocity in the rotating frame.

Then, to consider how accelerations are viewed in a rotating frame, put $\mathbf{B} = \mathbf{u}_{\text{I}}$ in Equation 4.2:

$$\begin{aligned}\left(\frac{d\mathbf{u}}{dt}\right)_{\text{I}} &= \left(\frac{d\mathbf{u}}{dt}\right)_{\text{R}} + \boldsymbol{\Omega} \times \mathbf{u}_{\text{I}} \\ &= \left(\frac{d}{dt}(\mathbf{u}_{\text{R}} + \boldsymbol{\Omega} \times \mathbf{X})\right)_{\text{R}} + \boldsymbol{\Omega} \times (\mathbf{u}_{\text{R}} + \boldsymbol{\Omega} \times \mathbf{X}) \\ &= \left(\frac{d\mathbf{u}}{dt}\right)_{\text{R}} + \boldsymbol{\Omega} \times \left(\frac{d\mathbf{X}}{dt}\right)_{\text{R}} + \boldsymbol{\Omega} \times \mathbf{u}_{\text{R}} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{X}) \\ &= \left(\frac{d\mathbf{u}}{dt}\right)_{\text{R}} + 2\boldsymbol{\Omega} \times \mathbf{u}_{\text{R}} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{X}).\end{aligned}$$

In an inertial frame Newton's law gives (for a fluid parcel of mass m)

$$\left(\frac{d\mathbf{u}}{dt}\right)_{\text{I}} = \frac{1}{m}\mathbf{F},$$

so when viewed in a rotating frame, the same parcel obeys

$$\left(\frac{d\mathbf{u}}{dt}\right)_{\text{R}} = \frac{1}{m}\mathbf{F} - 2\boldsymbol{\Omega} \times \mathbf{u}_{\text{R}} - \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{X}).$$

The extra “force” terms that appear on the right-hand-side are a result of working in a non-inertial reference frame. Since the forces are not real, they are often referred to as *fictitious* or *apparent* forces. The second term on the right-hand-side is known as the **Coriolis force**. If $\boldsymbol{\Omega}$ points up (down), the Coriolis force deflects horizontally moving objects to the right (left) when observed in a rotating frame of reference.

The third term on the right-hand-side is known as the **Centrifugal force**. This force appears to push objects away from the rotation axis.

Now that we can treat accelerations appropriately, we can write down the momentum equation in a rotating frame of reference. We just need to replace $\frac{D\mathbf{u}_I}{Dt}$ with

$$\frac{D\mathbf{u}_R}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{u}_R + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{X}). \quad (4.3)$$

For example, in the case of the Boussinesq equations, the momentum equation (2.6) becomes

$$\frac{D\mathbf{u}_R}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{u}_R + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{X}) = -\frac{1}{\bar{\rho}} \nabla p + b\mathbf{e}_z.$$

As density (or buoyancy) is a scalar, Equation 2.15 is unchanged in a rotating frame and therefore the Boussinesq equations in a rotating frame can be summarised as:

$$\frac{D\mathbf{u}}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{u} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{X}) = -\frac{1}{\bar{\rho}} \nabla p + b\mathbf{e}_z, \quad (4.4)$$

$$\nabla \cdot \mathbf{u} = 0, \quad (4.5)$$

$$\frac{Db}{Dt} + N^2(z)w = 0, \quad (4.6)$$

where we have dropped the subscript R from the velocity as we simply interpret $\mathbf{u}$ in these equations to be the velocity in a rotating frame of reference.

i Here we have allowed for the effects of rotation by writing the (non-hydrostatic) Boussinesq momentum equation in a rotating reference frame. Note that other equation sets (e.g., the shallow water, incompressible Navier-Stokes and hydrostatic Boussinesq equations) can be written in a rotating frame of reference in a similar way, i.e., by using Equation 4.3 to incorporate the Coriolis and centrifugal forces in the momentum equations. We will make use of this in future sections.

4.2.1 Centrifugal force as a potential

A simplification to these equations can be made if we are able to write the centrifugal force as the gradient of a scalar potential that can then be combined with the pressure. To do this, we define a set of cylindrical polar coordinates (s, θ, z) in which the z -axis is aligned with the rotation axis, i.e., with $\boldsymbol{\Omega}$. s is then the perpendicular distance from the z -axis. Since in cylindrical polars, the position vector is given by $\mathbf{X} = se_s + ze_z$ we have that

$$\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{X}) = \Omega \mathbf{e}_z \times \Omega s \mathbf{e}_\theta = -\Omega^2 s \mathbf{e}_s,$$

demonstrating that the centrifugal force $-\mathbf{\Omega} \times (\mathbf{\Omega} \times \mathbf{X})$ pushes objects away from the rotation axis. Now, we want to find a potential ϕ_c such that $\nabla\phi_c = \mathbf{\Omega} \times (\mathbf{\Omega} \times \mathbf{X})$. It follows that

$$\frac{\partial\phi_c}{\partial s} = -\Omega^2 s, \quad \frac{\partial\phi_c}{\partial\theta} = \frac{\partial\phi_c}{\partial z} = 0,$$

and hence $\phi_c = -\frac{\Omega^2 s^2}{2} = -\frac{1}{2}|\mathbf{\Omega} \times \mathbf{X}|^2$.

Writing the centrifugal force as $-\mathbf{\Omega} \times (\mathbf{\Omega} \times \mathbf{X}) = -\nabla\phi_c$ allows us to simplify Equation 4.4 to give

$$\frac{D\mathbf{u}}{Dt} + 2\mathbf{\Omega} \times \mathbf{u} = -\frac{1}{\rho}\nabla p_r + b\mathbf{e}_z, \quad (4.7)$$

where $p_r = p + \bar{\rho}\phi_c$.

i To simplify the notation we drop the subscript r in the rest of this course, while we consider the modified pressure that includes the centrifugal effect.

i Similar to centrifugal force, gravity can be formulated as a potential ϕ_g such that $\mathbf{g} = -\nabla\phi_g$. Taking the Earth to be a perfect sphere of radius $r_e = 6371\text{km}$ and mass $M = 5.98 \times 10^{24}\text{kg}$, Newton's law of gravitation gives $\phi_g = -GM/r$ where $G = 6.67 \times 10^{-11}\text{m}^3\text{kg}^{-1}\text{s}^{-2}$ is the gravitational constant. Then $\mathbf{g} = \nabla(GM/r) = -(GM/r^2)\mathbf{e}_r$ (points radially inwards). At the Earth's surface $|\mathbf{g}| = GM/r_e^2 = 9.82\text{ms}^{-2}$. If the height (depth) changes are small enough, we can assume that $\mathbf{g}$ is locally constant, which is what we consider for the rest of this course. However, if in an application dynamics takes place in a thick layer of fluid, the variation of $\mathbf{g}$ should be properly taken into account.

4.3 Tangent-plane approximation

To study the fluid dynamics of oceans and atmospheres, it is convenient to work in a frame of reference rotating with the planet. In this course, our model planet will be the Earth. The spherical geometry of Earth can influence the dynamics of flows and make studying them more involved. However, the sphericity of the Earth is not always as important as its rotation. In this case, it is more convenient to use a locally Cartesian model and introduce two approximations.

In spherical polar coordinates (r, θ, ϕ) where r is a radial coordinate, θ is the latitude (angle measured northwards from the equator) and ϕ is the longitude we have that

$$\mathbf{\Omega} = \Omega \sin \theta \mathbf{e}_r + \Omega \cos \theta \mathbf{e}_\theta. \quad (4.8)$$

That is, $\mathbf{\Omega}$ in its most general form has a radially (locally vertical) and a meridional (locally horizontal) component which both depend on latitude θ . However, if we limit the dynamics to a region where the changes in θ are small, we can make a set of simplifications. We define a plane tangent to the surface of the Earth at a latitude θ_0 and use a Cartesian system to define the motion (see figure ??). We take the z -axis to point in the direction of $-\nabla\Phi$ and write $-\nabla\Phi = g\mathbf{e}_z$ where g is taken to be constant over the depth of the (thin) layer. x is taken to point eastwards and y northwards. We define $y = 0$ at $\theta = \theta_0$.

Since the geopotential force is directed almost radially inwards, we take $\mathbf{e}_r \approx \mathbf{e}_z$ and $\mathbf{e}_\theta \approx \mathbf{e}_y$ at $\theta = \theta_0$ and we make the approximation

$$\mathbf{\Omega} = \Omega \sin \theta \mathbf{e}_r + \Omega \cos \theta \mathbf{e}_\theta \approx \Omega \sin \theta \mathbf{e}_z + \Omega \cos \theta \mathbf{e}_y.$$

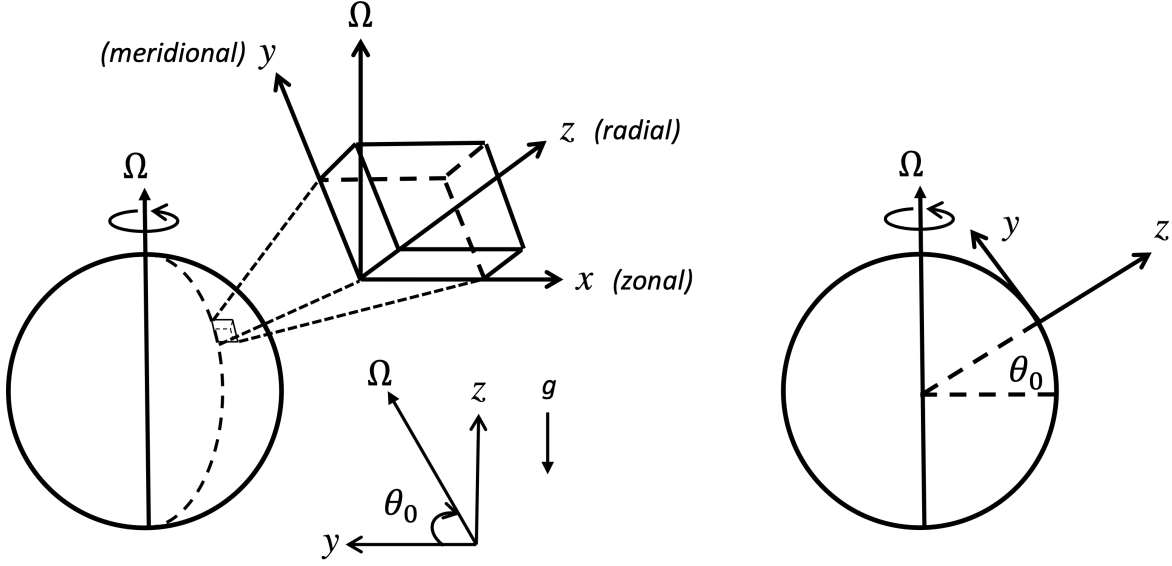


Figure 4.2: Tangent plane approximation.

For small $\theta - \theta_0$, $y \approx r_e(\theta - \theta_0)$ and so we have that

$$\begin{aligned}\sin \theta &= \sin(\theta_0 + (\theta - \theta_0)) = \sin \theta_0 + (\theta - \theta_0) \cos \theta_0 + O((\theta - \theta_0)^2) \approx \sin \theta_0 + \frac{\cos \theta_0}{r_e} y \\ \cos \theta &= \cos(\theta_0 + (\theta - \theta_0)) = \cos \theta_0 - (\theta - \theta_0) \sin \theta_0 + O((\theta - \theta_0)^2) \approx \cos \theta_0 - \frac{\sin \theta_0}{r_e} y.\end{aligned}$$

Hence we write $\boldsymbol{\Omega} = \Omega_z \mathbf{e}_z + \Omega_y \mathbf{e}_y$ where

$$\begin{aligned}\Omega_z &= \Omega \sin \theta_0 + \frac{\Omega \cos \theta_0}{r_e} y \\ \Omega_y &= \Omega \cos \theta_0 - \frac{\Omega \sin \theta_0}{r_e} y.\end{aligned}$$

The Coriolis force $2\boldsymbol{\Omega} \times \mathbf{u}$ can then be written as

$$2\boldsymbol{\Omega} \times \mathbf{u} = (2\Omega_y w - 2\Omega_z v, 2\Omega_z u, -2\Omega_y u)$$

Hence the Cartesian components of momentum equations under the tangent plane approximation are:

$$\frac{Du}{Dt} + 2(\Omega_y w - \Omega_z v) = -\frac{1}{\rho} \frac{\partial p}{\partial x} \quad (4.9)$$

$$\frac{Dv}{Dt} + 2\Omega_z u = -\frac{1}{\rho} \frac{\partial p}{\partial y} \quad (4.10)$$

$$\frac{Dw}{Dt} - 2\Omega_y u = -\frac{1}{\rho} \frac{\partial p}{\partial z} - g. \quad (4.11)$$

If we further assume that the layer is thin then the vertical flow speed is much smaller than the horizontal flow speeds, i.e., $|w| \ll |u|$ and $|v|$. We can therefore drop the $\Omega_y w$ term in Equation 4.9. Then, to ensure

energy is still conserved we must then also drop the $-\Omega_y u$ term from Equation 4.11 although in some ways this is less physically justified. This leads to the following set of equations:

$$\frac{Du}{Dt} - fv = -\frac{1}{\rho} \frac{\partial p}{\partial x} \quad (4.12)$$

$$\frac{Dv}{Dt} + fu = -\frac{1}{\rho} \frac{\partial p}{\partial y} \quad (4.13)$$

$$\frac{Dw}{Dt} = -\frac{1}{\rho} \frac{\partial p}{\partial z} - g, \quad (4.14)$$

where we have introduced $f = 2\Omega_z$. Equivalently, these equations can be written in vector form as

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\frac{1}{\rho} \nabla p - g\mathbf{e}_z, \quad (4.15)$$

where $\mathbf{f} = f(y)\mathbf{e}_z$. This formulation in which components of $\mathbf{\Omega}$ not in the direction of the local vertical is sometimes called the *traditional approximation*. Note, the centrifugal potential no longer appears in our equations, this is a result of defining the local vertical in the direction of $-\nabla\Phi$. Therefore, rotation enters the equations only through the Coriolis force and f is often called the *Coriolis parameter*.

f is an important parameter in rotating fluid dynamics and it is often written as

$$f = f_0 + \beta y, \quad \text{where } f_0 = 2\Omega \sin \theta_0, \quad \beta = \frac{2\Omega \cos \theta_0}{r_e}.$$

Note that f_0 is positive in the northern hemisphere ($\theta_0 > 0$), negative in the southern hemisphere ($\theta_0 < 0$), and zero at the equator ($\theta_0 = 0$). Representing f in this way is the basis of the so-called **β -plane approximation**, with the β term containing the latitudinal variations in f .

If the flows we care about are extremely localised then we do not need to take the latitudinal variations in f into account and we can make a further approximation given by $f = f_0 = 2\Omega \sin \theta_0$. This is known as the **f -plane approximation**. Note, under the f -plane approximation (i.e., with f constant), Equation 4.15 is analogous to that used in sections 4.4.1 - 4.4.3 where we took $\mathbf{\Omega} = \Omega\mathbf{e}_z$ (provided we replace 2Ω by f_0) and so physical concepts such as geostrophic balance, Taylor-Proudman, and thermal wind balance are relevant on the f -plane too.

Considering the tangent plane approximation $2\mathbf{\Omega} \approx \mathbf{f} = f\mathbf{e}_z$, we rewrite the important equations in geophysical fluid dynamics (Boussinesq and Shallow Water) that we focus on in this course in the rotating frame of reference.

Inviscid Shallow Water Equations

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} + \mathbf{f} \times \mathbf{v} = -g \nabla_H \eta. \quad (4.16)$$

$$\frac{\partial h}{\partial t} + \nabla_H \cdot (h\mathbf{v}) = 0, \quad (4.17)$$

$$h(x, y, t) = \eta(x, y, t) - \eta_b(x, y, t). \quad (4.18)$$

(Hydrostatic) Boussinesq Equations

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{v} + \mathbf{f} \times \mathbf{v} = -\frac{1}{\rho} \nabla_H p, \quad (4.19)$$

$$\frac{\partial p}{\partial z} = \bar{\rho}b = -\rho g, \quad (4.20)$$

$$\nabla \cdot \mathbf{u} = 0, \quad (4.21)$$

$$\frac{\partial b}{\partial t} + (\mathbf{u} \cdot \nabla)b + N^2 w = 0, \quad (4.22)$$

Rotating (Non-hydrostatic) Boussinesq Equations

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} + \mathbf{f} \times \mathbf{u} = -\frac{1}{\bar{\rho}}\nabla p + b\mathbf{e}_z, \quad (4.23)$$

$$\nabla \cdot \mathbf{u} = 0, \quad (4.24)$$

$$\frac{\partial b}{\partial t} + (\mathbf{u} \cdot \nabla)b + N^2(z)w = 0, \quad (4.25)$$

4.4 Rotationally dominated flow

4.4.1 Geostrophic balance

In some fluid flows (e.g. large-scale dynamics in the atmosphere and ocean), the Coriolis force and pressure gradients appear to be much larger than the other forces or acceleration terms. As a result, the horizontal components of Equation 4.23 (or Equation 4.19) reduce to

$$\mathbf{f} \times \mathbf{u} = -\frac{1}{\bar{\rho}}\nabla_H p. \quad (4.26)$$

This equation represents an important balance in geophysical fluid dynamics and is known as **geostrophic balance**. It follows from Equation 4.26 that $\mathbf{u} \cdot \nabla p = 0$ for the horizontal components of velocity. Then, since surfaces of constant pressure have normal ∇p , horizontal wind flows along lines of constant pressure.

i We will put the concept of geophysical balance on firmer mathematical footing in chapter 5.

Example

Consider a region of atmosphere rotating about a vertical axis ($\boldsymbol{\Omega} = \Omega \mathbf{e}_z$) which is at higher pressure than in surrounding regions (sometimes called a high pressure centre) so that the negative pressure gradient $-\nabla p$ is pointing away from the centre of the region. The gas will want to move from high pressure to low pressure but the gas will be pushed to the right by the Coriolis force so that it moves in a clockwise (or *anticyclonic*) motion around the high pressure centre. In the southern hemisphere anticyclones are still associated with regions of high pressure but in that case the motion will be anti-clockwise.

4.4.2 Taylor-Proudman theorem

If in the vertical direction we assume hydrostatic balance (as when making the hydrostatic approximation in Section 2.2.4) so that we have

$$\frac{\partial p}{\partial z} = -\rho g, \quad (4.27)$$

and if we consider the case where density is a function of z only and the background rotation is aligned with gravity, i.e., $\mathbf{\Omega} = \Omega \mathbf{e}_z$, then geostrophic balance in the horizontal direction gives us (see Equation 4.26)

$$-fv = -\frac{1}{\bar{\rho}} \frac{\partial p}{\partial x}, \quad fu = -\frac{1}{\bar{\rho}} \frac{\partial p}{\partial y}. \quad (4.28)$$

These equations can be rearranged to give

$$v = \frac{\partial}{\partial x} \frac{p}{f\bar{\rho}}, \quad u = -\frac{\partial}{\partial y} \frac{p}{f\bar{\rho}}. \quad (4.29)$$

It follows from here that $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$ and so it must be that $\frac{\partial w}{\partial z} = 0$ (from Equation 4.5). In other words, the vertical velocity is independent of height.

Now, since $\rho = \rho(z)$, Equation 4.27 gives $\frac{\partial^2 p}{\partial x \partial z} = \frac{\partial^2 p}{\partial y \partial z} = 0$ and so Equation 4.29 then gives us that $\frac{\partial u}{\partial z} = \frac{\partial v}{\partial z} = 0$ (recall $\bar{\rho}$ is a constant). Therefore we have that the horizontal flow is also independent of height.

We have shown what is commonly known as the **Taylor-Proudman theorem** which says within a fluid that is steadily rotated at high angular velocity, the fluid velocity will be uniform along any line parallel to the axis of rotation. For example, if there is a solid horizontal boundary, at the surface, say, then $w = 0$ at that boundary and so (since vertical derivatives are zero) $w = 0$ everywhere. Hence the flow is essentially two-dimensional.

i In reality, we do not see exact two-dimensional flow in the atmosphere or the oceans (essentially because of stratification). However, it is not uncommon in geophysical fluid dynamics for effects like Taylor-Proudman to manifest in the real flow, even if assumptions underlying a phenomena are not fully satisfied.

Example

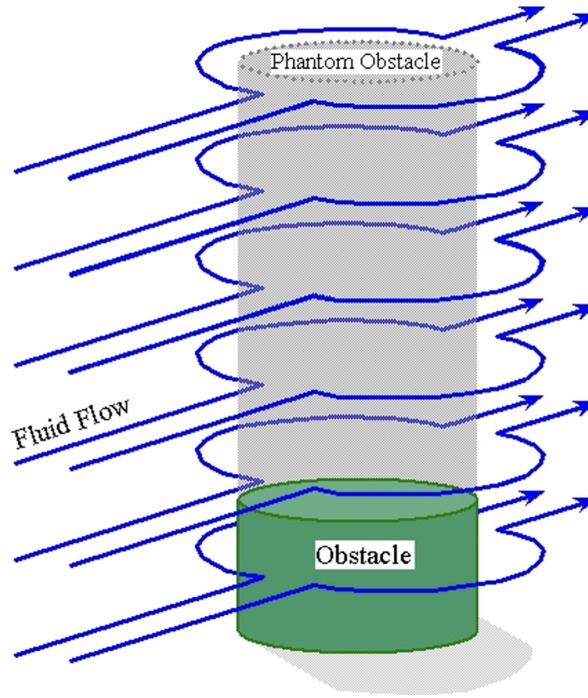


Figure 4.3: Taylor-Proudman column

If rotating fluid moves around an obstacle near the ground, the fluid must also move around an ‘imaginary’ obstacle higher up in the fluid. This is depicted in @fig-taylorproudman and can be observed in rotating tank experiments such as [this video](#).

4.4.3 Thermal wind balance

By combining geostrophic and hydrostatic approximations, we arrive at another important balance in geophysical fluid dynamics, so called **thermal wind balance**. To see the balance we simply differentiate the horizontal geostrophic balance equations (4.28) by z :

$$f \frac{\partial v}{\partial z} = \frac{1}{\bar{\rho}} \frac{\partial^2 p}{\partial z \partial x}, \quad f \frac{\partial u}{\partial z} = -\frac{1}{\bar{\rho}} \frac{\partial^2 p}{\partial z \partial y},$$

and combine these with hydrostatic balance from the hydrostatic Boussinesq equations (4.20) to give

$$f \frac{\partial v}{\partial z} = \frac{\partial b}{\partial x} \tag{4.30}$$

$$f \frac{\partial u}{\partial z} = -\frac{\partial b}{\partial y}. \tag{4.31}$$

These equations represent thermal wind balance and the vertical derivative of the geostrophic velocity (wind) is known as the thermal wind. If the buoyancy is constant (e.g., if the density is constant) then there is no shearing of the geostrophic wind and equations (4.30) - (4.31) reduce to the Taylor-Proudman constraint.

5 Geostrophic theory

5.1 Overview

Geophysical fluid dynamics embodies an incredibly wide range of spatial scales – from planetary scales of tens of thousand kilometers to millimeters where the effect of viscosity is felt or clouds and micro-organisms form. Such a wide spectrum adds to the complexity and the degrees of freedom in this system and makes the understanding of the dynamics very challenging. One of our primary tools in geophysical fluid dynamics is to narrow down our attention to a smaller span of scales, and then use relevant scaling to reduce the dynamics to something simpler that is easier to understand or solve. More specifically, if we consider the scale of thousand kilometers in the atmosphere (or hundreds of kilometers in the ocean), scaling arguments lead to **geostrophic theory**, which is one of the most celebrated results of geophysical fluid dynamics. The scales of thousand kilometers in the atmosphere (which are given a special name of **synoptic scales**) are the range of scales that characterise our daily weather. Hence, geostrophic theory has been the cornerstone of weather forecast for many decades. To investigate this theory, we start by introducing the appropriate scaling for different variables and then use asymptotic expansion to simplify the dynamics.

5.2 Scaling the equations and Rossby number

The goal of scaling is to understand the relative importance of different phenomena that influence the fluids motion. In geophysical fluids, we have seen that stratification and rotation play crucial roles in the dynamics. The characteristic length scale that marks the relative importance of these two phenomena is the *deformation radius* (also known as Rossby radius of deformation) defined as

$$\mathcal{L}_d = \frac{N\mathcal{H}}{f_0}, \quad (5.1)$$

where $\mathcal{H}$ is the vertical scale (f and N are Coriolis and Brunt–Väisälä frequencies introduced in previous chapters). Recall that stratification/buoyancy primarily acts in the vertical, whereas the leading order Coriolis force $f\mathbf{e}_z \times \mathbf{v}$ acts in the horizontal. Hence, $\mathcal{L}_d$ characterises the horizontal scale at which rotational effects become as important as buoyancy effects. In the atmosphere, the ratio N/f is typically of order 100, so for a vertical scale associated with the height of the tropopause (first layer of the atmosphere where the weather dynamics take place), $\mathcal{L}_d = 1000$ km. In synoptic or weather scales, we consider horizontal scales that close $\mathcal{L}_d$ or slightly larger. The deformation radius of the ocean is an order of magnitude smaller than that of the atmosphere because N/f is much smaller in the ocean. Hence, the order of hundred kilometer in the ocean is dynamically equivalent to thousand kilometer in the atmosphere.

We scale the flow variables with the following characteristic values

$$\begin{aligned} (x, y) = \mathcal{L}(x^*, y^*), \quad z = \mathcal{H}z^*, \quad t = \mathcal{T}t^*, \quad (u, v) = \mathcal{U}(u^*, v^*), \quad w = \mathcal{W}w^*, \\ p = \mathcal{P}p^*, \quad b = \mathcal{B}b^*, \quad \mathbf{f} = f_0\mathbf{f}^*, \end{aligned} \quad (5.2)$$

where $()^*$ are dimensionless. We also use $\nabla_H = 1/\mathcal{L} \nabla_H^*$, consistent with the scale of x and y . We then non-dimensionalise the (hydrostatic) Boussinesq equations

$$\frac{1}{f_0 \mathcal{T}} \frac{\partial \mathbf{v}^*}{\partial t^*} + \frac{\mathcal{U}}{f_0 \mathcal{L}} (\mathbf{v}^* \cdot \nabla_H^*) \mathbf{v}^* + \frac{\mathcal{W}}{f_0 \mathcal{H}} w^* \frac{\partial \mathbf{v}^*}{\partial z^*} + \mathbf{f}^* \times \mathbf{v}^* = -\frac{\mathcal{P}}{\bar{\rho} f_0 \mathcal{L} \mathcal{U}} \nabla_H^* p^*, \quad (5.3)$$

$$\frac{\bar{\rho} \mathcal{B} \mathcal{H}}{\mathcal{P}} b^* = \frac{\partial p^*}{\partial z^*}, \quad (5.4)$$

$$\nabla_H^* \cdot \mathbf{v}^* + \frac{\mathcal{W} \mathcal{L}}{\mathcal{U} \mathcal{H}} \frac{\partial w^*}{\partial z^*} = 0, \quad (5.5)$$

$$\frac{1}{f_0 \mathcal{T}} \frac{\partial b^*}{\partial t^*} + \frac{\mathcal{U}}{f_0 \mathcal{L}} (\mathbf{v}^* \cdot \nabla_H^*) b^* + \frac{\mathcal{W}}{f_0 \mathcal{H}} w^* \frac{\partial b^*}{\partial z^*} + \frac{N^2 \mathcal{W}}{\mathcal{B}} w^* = 0. \quad (5.6)$$

i We could have started from non-hydrostatic Boussinesq equations as well and by a scaling argument shows that the vertical momentum equation at the leading order is hydrostatic.

In Equations (5.3) - (5.6) one of the most important dimensionless parameters in geophysical fluid dynamics has surface

$$\text{Ro} = \frac{\mathcal{U}}{f_0 \mathcal{L}}, \quad (5.7)$$

which shows the ratio of inertial force to Coriolis force and is called **Rossby number**. If the Coriolis force is strong, we expect Ro to be small. Using the definition Rossby number (5.7), we rewrite (5.3) - (5.6) as

$$\frac{1}{f_0 \mathcal{T}} \frac{\partial \mathbf{v}^*}{\partial t^*} + \text{Ro} (\mathbf{v}^* \cdot \nabla_H^*) \mathbf{v}^* + \text{Ro} \frac{\mathcal{W} \mathcal{L}}{\mathcal{U} \mathcal{H}} w^* \frac{\partial \mathbf{v}^*}{\partial z^*} + \mathbf{f}^* \times \mathbf{v}^* = -\frac{\mathcal{P}}{\bar{\rho} f_0 \mathcal{L} \mathcal{U}} \nabla_H^* p^*, \quad (5.8)$$

$$\frac{\bar{\rho} \mathcal{B} \mathcal{H}}{\mathcal{P}} b^* = \frac{\partial p^*}{\partial z^*}, \quad (5.9)$$

$$\nabla_H^* \cdot \mathbf{v}^* + \frac{\mathcal{W} \mathcal{L}}{\mathcal{U} \mathcal{H}} \frac{\partial w^*}{\partial z^*} = 0, \quad (5.10)$$

$$\frac{1}{f_0 \mathcal{T}} \frac{\partial b^*}{\partial t^*} + \text{Ro} (\mathbf{v}^* \cdot \nabla_H^*) b^* + \text{Ro} \frac{\mathcal{W} \mathcal{L}}{\mathcal{U} \mathcal{H}} w^* \frac{\partial b^*}{\partial z^*} + \frac{N^2 \mathcal{W}}{\mathcal{B}} w^* = 0. \quad (5.11)$$

In the rest of this chapter we assume N to be constant. However, the geostrophic theory can be easily generalised to a variable N .

i Ro is named after the Swedish-American meteorologist Carl-Gustav Arvid Rossby, who first explain who first explained the large-scale motions of the atmosphere using fluid mechanics.

5.3 Leading order: geostrophic balance

Our goal in this section is to use the relevant scaling to the weather scale to simplify the Boussinesq equation. To that end, we consider the following assumptions

1. The Rossby number is small and we denote it by $\text{Ro} = \epsilon$ to emphasise this fact. This means the Coriolis effect is strong. The average wind speed in the atmosphere is about 8 m/s , the Coriolis parameter at mid-latitudes (about 35°) is 0.8×10^{-4} and as mentioned earlier we are considering the weather (synoptic) scales of $\mathcal{L} = 1000 \text{ km}$. With these values we get $\text{Ro} \approx 0.1$, which is indeed a small parameter. A similar calculation for the ocean scales of 100 km leads to even smaller Rossby numbers.

2. The horizontal scale of motion is not significantly larger than the deformation radius (Equation 5.1). More specifically,

$$\frac{\mathcal{L}_d}{\mathcal{L}} = O(1)$$

This is a direct translation of considering the synoptic scales (thousands of kilometers) in the atmosphere or the mesoscales (hundreds of kilometers) in the ocean.

3. The time scale is $\mathcal{T} = \mathcal{L}/\mathcal{U}$. This time scale sometimes is referred to as *eddy turnover* time, because if we consider an eddy or vortex with the velocity $\mathcal{U}$ and diameter $\mathcal{L}$, it will roughly take $\mathcal{L}/\mathcal{U}$ time to make a full turn. This time scale is also known as *inertial* or *advective* time scale, because it will make the inertial or advection term $\mathbf{v} \cdot \nabla \mathbf{v}$ the same order of magnitude as the time derivative. This assumption leads to $1/(\mathcal{T}f_0) = \text{Ro}$ in (5.8) - (5.11).
4. To keep $\partial w/\partial z$ in the continuity equation Equation 5.10, we assume the vertical velocity scales as $\mathcal{W} = \mathcal{U}\mathcal{H}/\mathcal{L}$. This is simply keeping the problem in more general form by allowing $\partial w/\partial z$ to exist at the leading order rather than imposing an extra assumption. In other words, such scaling allows $\mathcal{W} < \mathcal{U}\mathcal{H}/\mathcal{L}$ as the terms $\partial w/\partial z$ and $w\partial/\partial z$ may naturally become small or zero in the solution (our scaling does not prevent).
5. The beta effect is an order (ϵ) smaller than the f_0 . More precisely,

$$\mathbf{f}^* = \left(\frac{f_0 + \beta y^* \mathcal{L}}{f_0} \right) \mathbf{e}_z = \left(1 + \frac{\beta \mathcal{L}}{f_0} y^* \right) \mathbf{e}_z = (f_0^* + \epsilon \beta^* y^*) \mathbf{e}_z$$

where β^* is $O(1)$ due to the assumption made and $f_0^* = f_0/f_0 = 1$. Note that we still keep f_0^* in the equations (despite being simply 1) so the leading term of the Coriolis force will stand out in the non-dimensionalised equations. Likewise, we keep $\mathbf{f}_0^* = f_0^* \mathbf{e}_z$ instead of the unit vector $\mathbf{e}_z$.

Using the assumption 3, 4 and 5, we rewrite (5.8) - (5.11)

$$\epsilon \left(\frac{\partial \mathbf{v}^*}{\partial t^*} + (\mathbf{v}^* \cdot \nabla_H^*) \mathbf{v}^* + w^* \frac{\partial \mathbf{v}^*}{\partial z^*} \right) + (f_0^* + \epsilon \beta^* y^*) \mathbf{e}_z \times \mathbf{v}^* = -\frac{\mathcal{P}}{\bar{\rho} f_0 \mathcal{L} \mathcal{U}} \nabla_H^* p^*, \quad (5.12)$$

$$\frac{\bar{\rho} \mathcal{B} \mathcal{H}}{\mathcal{P}} b^* = \frac{\partial p^*}{\partial z^*}, \quad (5.13)$$

$$\nabla_H^* \cdot \mathbf{v}^* + \frac{\partial w^*}{\partial z^*} = 0, \quad (5.14)$$

$$\epsilon \left(\frac{\partial b^*}{\partial t^*} + (\mathbf{v}^* \cdot \nabla_H^*) b^* + w^* \frac{\partial b^*}{\partial z^*} \right) + \frac{N^2 \mathcal{W}}{\mathcal{B}} w^* = 0. \quad (5.15)$$

If we consider the limit of small ϵ (following assumption 1), in the horizontal momentum Equation 5.12 the Coriolis force will be larger than all other terms in the LHS. We know the Coriolis force is not small at the weather scale as the horizontal wind $\mathbf{v}$ is not negligible. Hence, the pressure gradient in the RHS has to be the same order as Coriolis force to balance it out. This require the following scaling for pressure

$$\mathcal{P} \sim \bar{\rho} f_0 \mathcal{L} \mathcal{U}$$

Following this scaling, we can move to the vertical momentum Equation 5.13 and infer the appropriate scaling for buoyancy

$$\mathcal{B} \sim \frac{f_0 \mathcal{L} \mathcal{U}}{\mathcal{H}}$$

After replacing for $\mathcal{B}$ and $\mathcal{P}$ all the pieces of scaling puzzle are complete and we have a set of equations for which we know the relative importance of all terms \begin{align}

$$\epsilon \left(\frac{\partial \mathbf{v}^*}{\partial t^*} + (\mathbf{v}^* \cdot \nabla_H^*) \mathbf{v}^* + w^* \frac{\partial \mathbf{v}^*}{\partial z^*} \right) + \mathbf{f}_0^* \times \mathbf{v}^* + \epsilon \beta^* y^* \mathbf{e}_z \times \mathbf{v}^* = -\nabla_H^* p^*, \quad (5.16)$$

$$b^* = \frac{\partial p^*}{\partial z^*}, \quad (5.17)$$

$$\nabla_H^* \cdot \mathbf{v}^* + \frac{\partial w^*}{\partial z^*} = 0, \quad (5.18)$$

$$\epsilon \left(\frac{\partial b^*}{\partial t^*} + (\mathbf{v}^* \cdot \nabla_H^*) b^* + w^* \frac{\partial b^*}{\partial z^*} \right) + \left(\frac{\mathcal{L}_d}{\mathcal{L}} \right)^2 w^* = 0. \quad (5.19)$$

where we have used the definition of deformation radius.

In the next step, we start with the classical asymptotic approach to reduce the above equations to a simpler set. We expand all variables in terms of ϵ

$$\mathbf{v}^* = \mathbf{v}_0^* + \epsilon \mathbf{v}_1^* + \dots, \quad w^* = w_0^* + \epsilon w_1^* + \dots, \quad p^* = p_0^* + \epsilon p_1^* + \dots, \quad b^* = b_0^* + \epsilon b_1^* + \dots$$

Upon substitution in Equation 5.16 and keeping the leading order terms (the smallest power of ϵ), we find

$$\mathbf{f}_0^* \times \mathbf{v}_0^* = -\nabla_H p_0^*, \quad (5.20)$$

This is the infamous **geostrophic balance**, which is the building block of modern weather forecast. It expresses that the Coriolis forces is balancing the pressure gradient. If we take the divergence of both sides in Equation 5.20, we find

$$\nabla_H^* \cdot \mathbf{v}_0^* = 0, \quad (5.21)$$

which tells us that the horizontal velocity is divergence-free at the leading order. As a result, we can define the streamfunction below

$$u_0^* = -\frac{\partial \psi^*}{\partial y^*}, \quad v_0^* = \frac{\partial \psi^*}{\partial x^*}. \quad (5.22)$$

This looks similar to the definition of streamfunction for 2D incompressible flow. However, unlike 2D flow, ψ^* , u_0^* and v_0^* can depend on z . Considering Equation 5.20, we can set

$$\psi^* = p_0^* / f_0,$$

which automatically satisfies Equation 5.22. At leading order, Equation 5.18 becomes

$$b_0^* = \frac{\partial p_0^*}{\partial z} \quad (5.23)$$

This is simply the **hydrostatic balance**. Consider the conservation of mass at leading order

$$\nabla_H^* \cdot \mathbf{v}_0^* + \frac{\partial w_0^*}{\partial z^*} = 0.$$

We have already established that $\nabla_H^* \cdot \mathbf{v}_0^* = 0$ in Equation 5.21. Hence,

$$\frac{\partial w_0^*}{\partial z^*} = 0, \quad (5.24)$$

implying that if w_0^* is zero somewhere (like at the boundary) then it is zero everywhere.

Considering the assumption 2, the buoyancy Equation 5.19 at leading order is

$$\left(\frac{\mathcal{L}_d}{\mathcal{L}} \right)^2 w_0^* = 0, \quad (5.25)$$

showing that the leading order the vertical velocity is zero.

i Note that Equation 5.25 is in a sense stronger than Equation 5.24, but it requires $\mathcal{L}_d/\mathcal{L} = O(1)$. For $\mathcal{L} > \mathcal{L}_d$, all the leading order results hold (geostrophic and hydrostatic balance), except Equation 5.25. This range scale correspond to planetary scales.

As discussed earlier in Section 4.4.3, we can also combined hydrostatic and geostrophic balance by taking the z -derivative of Equation 5.20 and substitute for pressure terms from Equation 5.23

$$\mathbf{f}_0^* \times \frac{\partial \mathbf{v}_0^*}{\partial z^*} = -\nabla_H b_0^* \quad (5.26)$$

which is the *thermal wind balance* that holds at leading order.

We take a short break here from asymptotics and look at the implications and applications of geostrophic balance before proceeding to the next order.

5.3.1 Geostrophic Balance and weather forecast

Equation 5.20 is one of the most important equations in the large scale atmospheric physics and weather forecast. For example by knowing the pressure distribution, we can find the geostrophic velocity using this equation. $-\nabla_H p_0$ in meteorology is known as *pressure gradient force* (PGF), the direction of which is from high to low pressure. This can be slightly confusing as the pressure gradient itself $\nabla_H p_0$ is in the opposite direction to PGF, but consider the force that fluid parcel feels is from high to low pressure. Knowing PGF and using cross product rule in Equation 5.20, we find that if $\mathbf{f}$ points upward (or out of the horizontal plane), when we face the direction of PGF, the geostrophic velocity should be to our right. This is the case in the Northern hemisphere where $\mathbf{f}$ points upward. The situation is reversed in the Southern hemisphere as $\mathbf{f}$ points downward. For instance, if the PGF is pointing northward in the Southern hemisphere, the geostrophic wind blows toward the west. Following this argument, we also find that the geostrophic velocity is aligned with constant-pressure contours known as *isobar*. As illustrated in the example offigure Figure 5.3, using the direction of PGF we can find the direction of geostrophic velocity on weather maps (pressure contours)

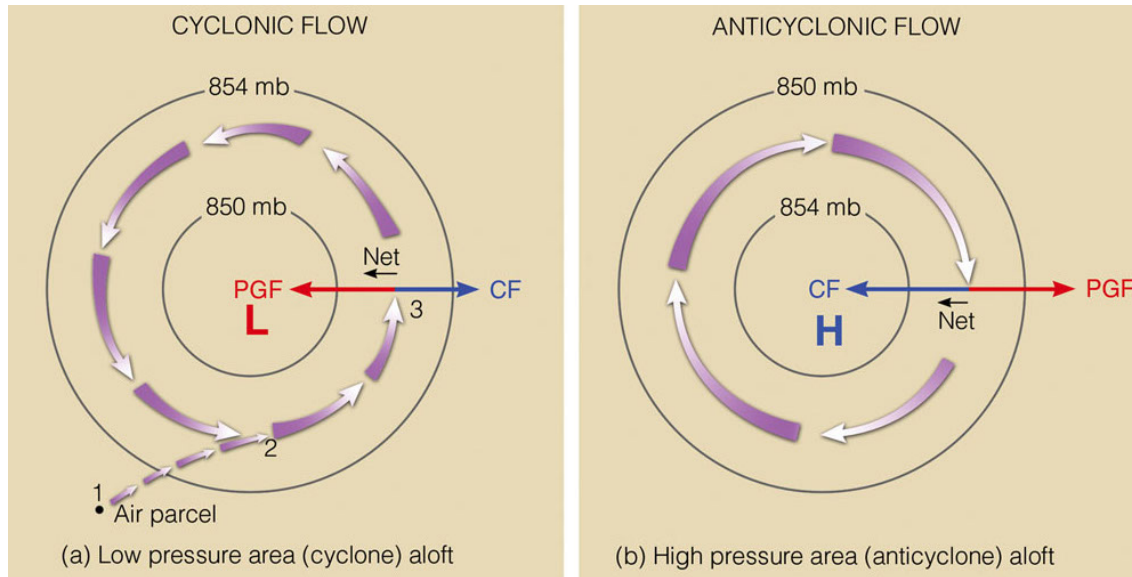


Figure 5.1: The direction of motion caused by geostrophic wind in cyclone (left) and anticyclone (right) in the Northern hemisphere.

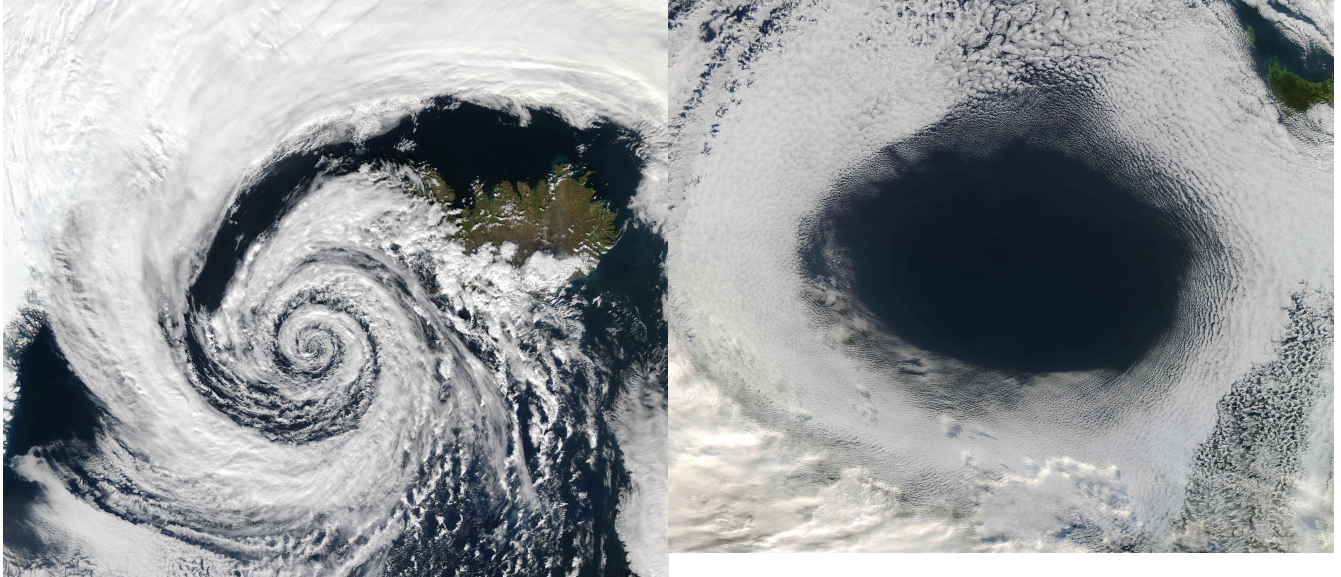


Figure 5.2: Satellite images of a cyclone (low-pressure system) in the left panel and an anticyclone (high-pressure system) in the right panel.

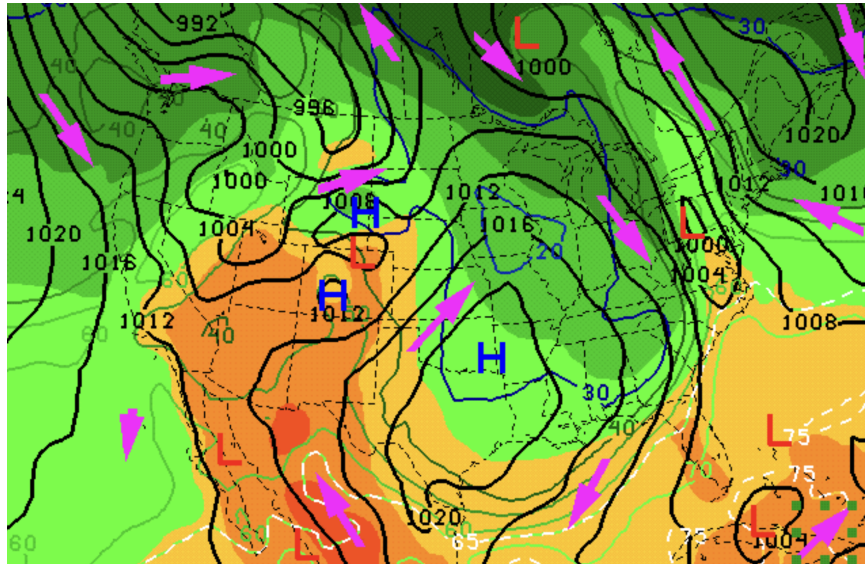


Figure 5.3: Pressure contours (isobars) on a weather map (in Northern hemisphere). The magenta arrows show the direction of geostrophic velocity.

The geostrophic balance is also fundamental in understanding the pressure systems. These systems are basically giant vortices that form our weather. For the moment, let's consider the Northern hemisphere. In a high pressure system, the pressure increase as we move toward the centre (see figure). Hence, PGF points outward. As a result, the fluid parcels should rotate clockwise in this system as shown in Figure 5.1 (following our argument in the preceding paragraph). This system is known as *anticyclone*. In contrast, the direction of motion is anticlockwise in a low-pressure system (see Figure 5.1), which is known as *cyclone*. With a similar reasoning we find out that cyclones (low-pressure systems) move anticlockwise and anticyclones (high-pressure systems) move clockwise in the Southern hemisphere. When a cyclone moves over sea, it absorbs moisture into its core as it has low pressure compared to its surrounding. This moisture turns into cloud and is carried in-land and create rainy weather. Hence, low-pressure systems (cyclones) are associated with rain and clouds and high-pressure systems with sunny weather. Such characteristics of these systems are beautifully capture in satellite images such as those shown in Figure 5.2. The clouds are attracted toward the centre of cyclone, whereas they are repelled from the centre of anticyclone.

i The thermodynamics behind the formation of cyclones/anticyclones is beyond the scope of this course but is an important area of study in meteorology. More particularly, development or strengthening of cyclones in the atmosphere can lead to the formation of hurricanes (this branch of science is so important that is given a name: *Cyclogenesis*).

5.4 Next order: quasigeostrophy

The leading order asymptotics revealed very useful information about the geostrophic balance and vertical velocity. However, these relations are *diagnostic* and time independent. In other words, they show, for example, how velocity is related to pressure but they do not show how velocity or pressure evolve in time. We know that the large scale atmosphere and ocean are dynamic and change in time. Hence, to find the time evolution of the flow variables we have to go to the next order (i.e. $O(\epsilon)$).

Starting with horizontal momentum equation Equation 5.16, we find

$$\frac{\partial \mathbf{v}_0^*}{\partial t^*} + (\mathbf{v}_0^* \cdot \nabla_H^*) \mathbf{v}_0^* + \beta^* y^* \mathbf{e}_z \times \mathbf{v}_0^* + \mathbf{f}_0^* \times \mathbf{v}_1^* = -\nabla_H^* p_1^*.$$

We then take the curl of this equation

$$\begin{aligned} \frac{\partial}{\partial x^*} \left(\frac{\partial v_0^*}{\partial t^*} + u_0^* \frac{\partial v_0^*}{\partial x^*} + v_0^* \frac{\partial v_0^*}{\partial y^*} + \beta^* y^* u_0 + f_0^* u_1 \right) - \frac{\partial}{\partial y^*} \left(\frac{\partial u_0^*}{\partial t^*} + u_0^* \frac{\partial u_0^*}{\partial x^*} + v_0^* \frac{\partial u_0^*}{\partial y^*} - \beta^* y^* v_0 - f_0^* v_1 \right) &= 0 \\ \rightarrow \frac{\partial \zeta_0^*}{\partial t^*} + (\mathbf{v}_0^* \cdot \nabla_H^*) \zeta_0^* + \beta^* v_0 &= -f_0^* \nabla_H^* \cdot \mathbf{v}_1^*, \end{aligned}$$

where we used Equation 5.21 and $\zeta_0^* = \partial v_0^* / \partial x^* - \partial u_0^* / \partial y^*$ is the vertical vorticity at zeroth order. The next order mass conservation equation is

$$\nabla_H^* \cdot \mathbf{v}_1^* + \frac{\partial w_1^*}{\partial z^*} = 0,$$

which can be used to replace for $\nabla_H^* \cdot \mathbf{v}_1^*$ in @q-QGvort1

$$\frac{\partial \zeta_0^*}{\partial t^*} + (\mathbf{v}_0^* \cdot \nabla_H^*) \zeta_0^* + \beta^* v_0 = f_0^* \frac{\partial w_1^*}{\partial z^*} \quad (5.27)$$

At next order, we obtain an evolution equation for the buoyancy as well

$$\frac{\partial b_0^*}{\partial t^*} + (\mathbf{v}_0^* \cdot \nabla_H^*) b_0^* + \left(\frac{\mathcal{L}_d}{\mathcal{L}} \right)^2 w_1^* = 0.$$

We take the z derivative of this equation and use it to substitute for $\partial w_1^*/\partial z^*$ in the RHS of Equation 5.27

$$\frac{\partial \zeta_0^*}{\partial t^*} + (\mathbf{v}_0^* \cdot \nabla_H^*) \zeta_0^* + \beta^* v_0 = f_0^* \left(\frac{\mathcal{L}}{\mathcal{L}_d} \right)^2 \frac{\partial}{\partial z^*} \left(\frac{\partial b_0^*}{\partial t^*} + (\mathbf{v}_0^* \cdot \nabla_H^*) b_0^* \right) \quad (5.28)$$

After expanding the RHS of this equation

$$\frac{\partial}{\partial z^*} \left(\frac{\partial b_0^*}{\partial t^*} + \mathbf{v}_0^* \cdot \nabla_H^* b_0^* \right) = \frac{\partial}{\partial t^*} \frac{\partial b_0^*}{\partial z^*} + \mathbf{v}_0^* \cdot \nabla_H^* \frac{\partial b_0^*}{\partial z^*} + \frac{\partial \mathbf{v}_0^*}{\partial z^*} \cdot \nabla_H^* b_0^*,$$

we find the last term is zero, because according to Equation 5.26 $\nabla_H^* b_0^*$ and $\partial \mathbf{v}_0^*/\partial z^*$ are normal to each other. Hence, Equation 5.28 simplifies to

$$\frac{D_0^*}{Dt} (\zeta_0^* + f^*) = f_0^* \left(\frac{\mathcal{L}}{\mathcal{L}_d} \right)^2 \frac{D_0^*}{Dt} \frac{\partial b_0^*}{\partial z^*}, \quad (5.29)$$

where

$$\frac{D_0^*}{Dt} = \frac{\partial}{\partial t^*} + \mathbf{v}_0^* \cdot \nabla_H^*.$$

We can rewrite Equation 5.29 as a material conservation of single quantity

$$\frac{D_0^* q^*}{Dt} = 0, \quad \text{where} \quad q^* = \zeta_0^* + f^* + f_0^* \left(\frac{\mathcal{L}}{\mathcal{L}_d} \right)^2 \frac{\partial b_0^*}{\partial z^*}. \quad (5.30)$$

This is the quasi-geostrophic (QG) balance in the non-dimensional form. We can transform all variables back to their dimensional version (using Equation 5.2) and drop the subscripts assuming that the flow is approximated by its leading order terms to arrive at

$$\frac{Dq}{Dt} = 0, \quad \text{where} \quad q = \zeta + f + \frac{1}{N^2} \frac{\partial b}{\partial z}. \quad (5.31)$$

which is the QG equation in the dimensional form. q in this equation is the known as **potential vorticity** (PV), which is of immense significance in geophysical fluid dynamics. Equation 5.31 implies that following each particle in a QG flow we find its PV remains unchanged. Note that all the variables in Equation 5.31 can be written in terms of a streamfunction ψ (the dimensional version of ψ^* in Equation 5.22):

- horizontal velocity components: $u = -\partial\psi/\partial y$ and $v = \partial\psi/\partial x$,
- vertical vorticity (vorticity in the z direction): $\zeta = \nabla_H^2 \psi$,
- perturbation buoyancy: $b = f_0 \partial\psi/\partial z$.

This means that QG equation is basically a partial differential equation for only one variable. Considering that the full Boussinesq equations contain 5 unknown variables, this is a significant simplification to describe the large scale atmospheric and oceanic flows (thanks to mathematics and in particular perturbation methods!).

5.5 Shallow water quasigeostrophy

We can follow a similar procedure for the rotating shallow water system. At leading order, we find the geostrophic balance in the form of

$$\mathbf{f} \times \mathbf{u} = -g \nabla_H \eta$$

At next order, we find a similar material conservation

$$\frac{Dq}{Dt} = 0, \quad \text{where} \quad q = \zeta + f + -\frac{f_0}{H}\eta = \nabla_H^2\psi + f - \frac{1}{\mathcal{L}_d^2}\psi. \quad (5.32)$$

which is the QG equation for shallow waters. This form of q is the shallow water PV and

$$\mathcal{L}_d = \frac{\sqrt{gH}}{f_0} \quad (5.33)$$

is the deformation radius for this system.

i The derivation of shallow water quasigeostrophy is rather shorter and simpler than that of the Boussinesq system detailed above. We leave this for you in the assignments.

6 Waves in the atmosphere and ocean

6.1 Basic concepts

We review some basic concepts about waves that you might already be familiar with from other courses. To start, consider a plane sinusoidal wave

$$\psi(\mathbf{x}, t) = \mathcal{M} \cos(\mathbf{k} \cdot \mathbf{x} - \omega t) + \mathcal{N} \sin(\mathbf{k} \cdot \mathbf{x} - \omega t), \quad (6.1)$$

where $\mathbf{k} = (k_x, k_y, k_z)$ is the wavevector and ω the frequency. The norm of wavevector $k = |\mathbf{k}|$, is known as the wavenumber (in the literature even $\mathbf{k}$ is sometimes loosely referred to as the wavenumber). The argument of sin and cos functions is known as the phase and usually denoted by $\theta = \mathbf{k} \cdot \mathbf{x} - \omega t$. Using trigonometric identities, we can write Equation 6.1 in the equivalent different form

$$\psi(\mathbf{x}, t) = \mathcal{A} \cos(\mathbf{k} \cdot \mathbf{x} - \omega t + \phi_0),$$

where $\mathcal{A} = \sqrt{\mathcal{M}^2 + \mathcal{N}^2}$ is the wave amplitude and ϕ_0 is an initial phase shift.

Another useful equivalent form of Equation 6.1 is in terms of the complex exponential function:

$$\psi(\mathbf{x}, t) = \mathcal{R}e\{Ae^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}\} \quad (6.2)$$

where A is the complex wave amplitude (often loosely referred to simply as the wave amplitude). For plane waves A is constant, but in general it may depend on space and time. With some complex algebra one can show that $|A| = \mathcal{A}$.

In the remainder of this chapter, we will often omit the real-part operator $\mathcal{R}e$ to lighten the notation, with the understanding that the physical solution is always the real part of the complex expression.

From @planewave_sin we can infer that the distance between two consecutive crests (or troughs) is

$$\lambda = \frac{2\pi}{|\mathbf{k}|},$$

which is called the wavelength. If we fix a point in space, the period

$$T = \frac{2\pi}{\omega}$$

is the time it takes for the wave to repeat itself at that point (for example, the time between two crests passing a fixed location).

6.1.1 Dispersion relation

Different kinds of waves arise as solutions of different linear (partial) differential equations (there are also nonlinear waves, which are beyond the scope of this course). For example, the plane wave considered above (in one spatial dimension) can be a solution of

$$\frac{\partial \psi}{\partial t} + \alpha \frac{\partial \psi}{\partial x} = 0. \quad (6.3)$$

Upon substituting Equation 6.2 into this equation, we obtain $A(-i\omega + \alpha ik_x) = 0$. For a non-trivial solution ($A \neq 0$), this requires

$$\omega = \alpha k_x \quad (6.4)$$

In other words, for any wavenumber k_x we can find a corresponding frequency ω from Equation 6.4 such that $Ae^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$ is a solution of Equation 6.3. A relation between ω and $\mathbf{k}$ obtained in this way is called the *dispersion relation*.

Example

For the following equation

$$\frac{\partial}{\partial t} \nabla^2 \psi + \beta \frac{\partial \psi}{\partial x} = 0, \quad (6.5)$$

we find the dispersion relation

$$\omega = \frac{-\beta k_x}{k_x^2 + k_y^2 + k_z^2} \quad (6.6)$$

For a general linear equation, the dispersion relation can have the form of

$$\omega = \Omega(\mathbf{k}; \mathbf{x}, k)$$

6.1.2 Phase speed and phase velocity

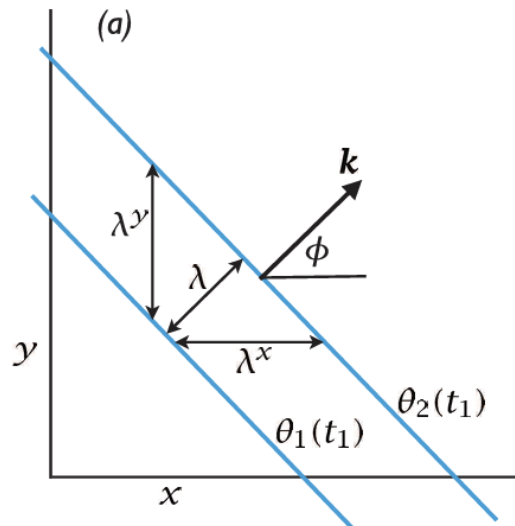


Figure 6.1: Propagation of plane wave crests.

Phase speed is the distance that the crests (or troughs) of a wave are travelling per unit of time. To find the value of phase speed, (at any instance in time) we align our coordinate along the $\mathbf{k}$ axis as shown in Figure 6.1. If denote the position along the $\mathbf{k}$ axis in the new coordinate by x^* , $\mathbf{k} \cdot \mathbf{x} = kx^*$. We then write Equation 6.2 as

$$\psi = Ae^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} = Ae^{ik(x^* - ct)},$$

where

$$c_p = \frac{\omega}{k} \quad (6.7)$$

is the phase speed. From this form we can see that the wave function is simply shifted along the $\mathbf{k}$ direction by ct . Hence, if we pick a point on the wave (for example on a crest), it moves along the $\mathbf{k}$ axis with the speed c .

Example

Based on the definition of phase speed in Equation 6.7, we derive c for the following equations

$$\frac{\partial \psi}{\partial t} + \alpha \frac{\partial \psi}{\partial x} = 0 \quad (6.8)$$

$$\frac{\partial}{\partial t} \frac{\partial^2 \psi}{\partial x^2} + \beta \frac{\partial \psi}{\partial x} = 0. \quad (6.9)$$

We can first derive the dispersion relation for the above systems and then divide ω by k to get the phase speed. We have already found the dispersion for Equation 6.8 in Equation 6.4. Dividing this expression by k , we obtain $c_p = \alpha$. The system Equation 6.9 is a one-dimensional case of Equation 6.5, where $k = k_x$ and $k_y = k_z = 0$. Dividing Equation 6.6 by k leads to $c_p = -\beta/k_x$.

It is common to assign a direction to the phase speed and turn it into a velocity vector. Considering that wave crests propagate along $\mathbf{k}$ we can define the phase velocity as

$$\mathbf{c}_p = \frac{\omega}{k} \frac{\mathbf{k}}{k}.$$

(decide later if you want to talk about phase velocity in more detail. probably better not to bother)

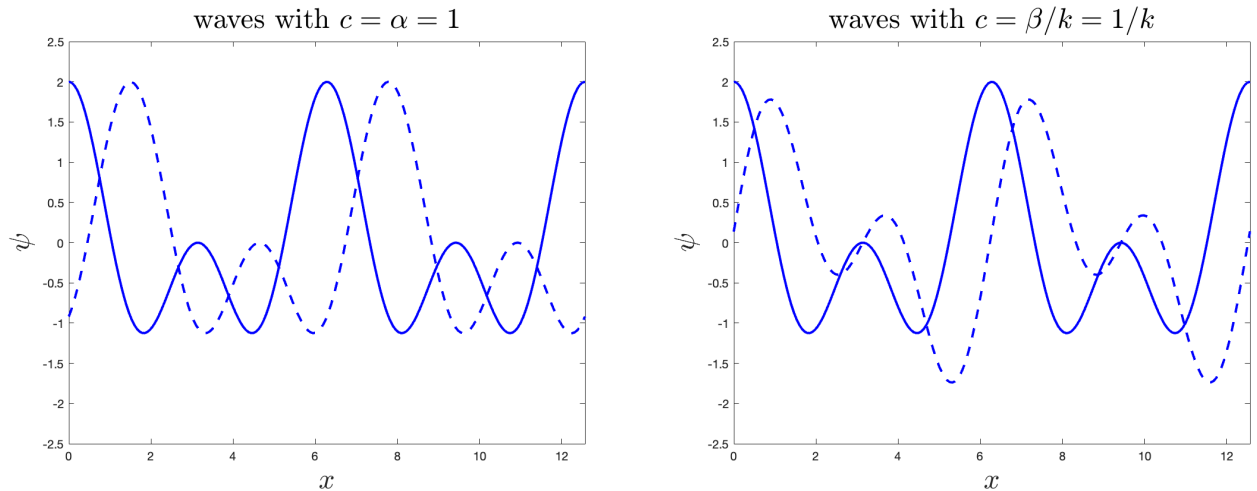


Figure 6.2: Propagation of two superimposed waves: left) non-dispersive and right) dispersive. The solid line is the initial wave $\psi(x, t = 0)$ and the dashed line after 1.5 time $\psi(x, t = 1.5)$.

Now let's consider a (one-dimensional) compound wave that consists of two wavenumbers, for example

$$\psi = A_1 e^{i(x-c_1 t)} + A_2 e^{i2(x-c_2 t)}. \quad (6.10)$$

If this wave evolves under Equation 6.8 we find that phase velocity of both components are the same $c_1 = c_2 = \alpha$. This means that both signals (wave components) move at the same speed so the initial shape of the wave will be preserved in time and just gets shifted along the x axis. The story is very different for the system Equation 6.9 as each component of the wave has a different velocity $c_1 = \beta$ and $c_2 = \beta/2$. This means that the shape of the wave deforms in time as its constituents propagate at different speed. This is demonstrated in figure Figure 6.2. The first systems Equation 6.8 belongs to a class of waves that are called *non-dispersive*. For non-dispersive waves the phase speed does not depend on the wavenumber (left panel of Figure 6.2). The second system Equation 6.9 is an example of *dispersive* waves for which the phase speed is an explicit function of wavenumber (right panel of Figure 6.2).

6.1.3 Group velocity

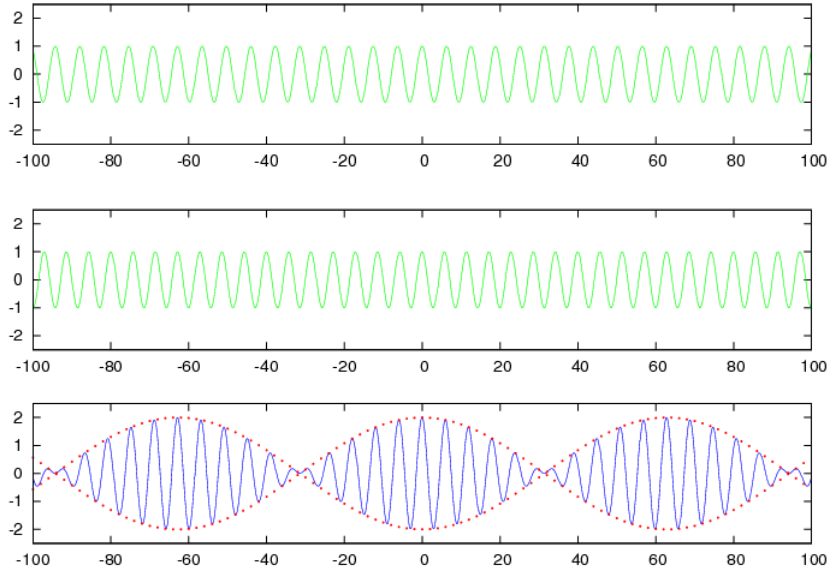


Figure 6.3: The signals of first and second row plots, with slightly different frequencies are added generating the bottom plot signal.

Energy and information do not necessarily travel at the phase speed. We do not care too much about how individual crests and troughs propagate in time. Instead, we want to know how a packet or bundle of them move together. Their collective motion shows how much energy and information is transferred by them. The quantity that characterises such a collective motion is the *group velocity*.

To understand the concept of group velocity, we consider a superposition of two waves similar to Equation 6.10

$$\psi = A \left[e^{i(k_1 x - \omega_1 t)} + e^{i(k_2 x - \omega_2 t)} \right], \quad (6.11)$$

but this time we set the frequency of individual signals to $\omega_1 = \omega + \Delta\omega$ and $\omega_2 = \omega - \Delta\omega$, and their wavenumbers to $k_1 = k + \Delta k$ and $k_2 = k - \Delta k$. If Δk and $\Delta\omega$ are small, these two components will have similar frequencies and wavenumbers. An example of two wave signals with slightly different frequencies and wavenumbers is depicted in the top rows of Figure 6.3. The bottom row of this figure shows the

superposition of these waves, which forms a wave packets (an envelope of crests and troughs) that look like a wave itself. We can see this in a mathematical expression by simplifying Equation 6.11

$$\psi = Ae^{i(kx-\omega t)} \left[e^{i(\Delta kx-\Delta\omega t)} + e^{-i(\Delta kx-\Delta\omega t)} \right] = A \cos(\Delta kx - \Delta\omega t) e^{i(kx-\omega t)}$$

The amplitude of the superposition is no longer constant and instead is *modulated* by $\cos(\Delta kx - \Delta\omega t)$. Just like the original wave signals, this modulated envelope moves in time, however, with a different speed that is $\Delta\omega/\Delta k$. This quantity at the limit of $\Delta k, \Delta\omega \rightarrow 0$ is the group velocity $c_g = \partial\omega/\partial k$. Group velocity is equal to the phase speed only for non-dispersive waves, i.e. when the frequency is a linear function of wavenumber. The energy in a wave is transported by group velocity. To have a better understanding of this phenomenon, consider the nodes between the envelopes in figure Figure 6.3. The wave function is zero at these nodes so the energy cannot be transferred across them. These nodes (as well as the envelopes between them that contains the wave energy) move with the group velocity. We can generalise this concept to three dimensions and define the group velocity as

$$\mathbf{c}_g = \nabla_{\mathbf{k}} \omega = \frac{\partial \omega}{\partial \mathbf{k}} = \left(\frac{\partial \omega}{\partial k_x}, \frac{\partial \omega}{\partial k_y}, \frac{\partial \omega}{\partial k_z} \right) \quad (6.12)$$

6.2 Poincaré waves in the Shallow Water equations

Now that we are equipped with some basic concepts from wave dynamics, we can study waves in geophysical flows. One might ask why waves appear so prominently in large-scale geophysical systems. In the previous chapter we saw that the leading-order dynamics at large scales ($\mathcal{L} \approx \mathcal{L}_d$) is quasigeostrophic. With a different timescale, and still at small Rossby number with $\mathcal{L} \approx \mathcal{L}_d$, wave dynamics can also emerge. In this course we demonstrate this for the Shallow Water system; similar arguments can be made for the Boussinesq equations.

We consider the Shallow Water equations with a flat bottom and place $z = 0$ at the mean free-surface level. With these assumptions, $\eta_b = -H$ and $h = \eta + H$, where H is the (constant) mean depth. Writing the Shallow Water equations in terms of η and $\mathbf{v}$ (substituting $h = \eta + H$) gives

$$\begin{aligned} \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla_H) \mathbf{v} + \mathbf{f} \times \mathbf{v} &= -g \nabla_H \eta, \\ \frac{\partial \eta}{\partial t} + H \nabla_H \cdot \mathbf{v} + \nabla_H \cdot (\eta \mathbf{v}) &= 0. \end{aligned} \quad (6.13)$$

This formulation allows us to distinguish the scale of height variations from the mean depth H . We now non-dimensionalise Equation 6.13 using

$$(x, y) = \mathcal{L}(x^*, y^*), \quad (u, v) = \mathcal{U}(u^*, v^*), \quad t = \frac{1}{f_0} t^*, \quad \eta = \frac{f_0 \mathcal{L} \mathcal{U}}{g} \eta^*, \quad \mathbf{f} = f_0 \mathbf{f}^*. \quad (6.14)$$

This is very similar to the scaling we used for geostrophic theory, except that we now choose the characteristic time scale to be $1/f_0$ rather than the advective time scale $\mathcal{L}/\mathcal{U}$. In large-scale geophysical flows, $1/f_0$ is typically shorter than $\mathcal{L}/\mathcal{U}$, so we are focusing on dynamics that are faster than quasigeostrophic motions.

The dimensionless shallow water equations obtained from Equation 6.14 are

$$\begin{aligned} \frac{\partial \mathbf{v}^*}{\partial t^*} + \text{Ro} (\mathbf{v}^* \cdot \nabla_H^*) \mathbf{v}^* + \mathbf{f}^* \times \mathbf{v}^* &= -\nabla_H^* \eta^*, \\ \frac{\partial \eta^*}{\partial t^*} + \left(\frac{\mathcal{L}_d}{\mathcal{L}} \right)^2 \nabla_H^* \cdot \mathbf{v}^* + \text{Ro} \nabla_H^* \cdot (\eta^* \mathbf{v}^*) &= 0, \end{aligned} \quad (6.15)$$

where $\mathcal{L}_d = \sqrt{gH}/f_0$ is the deformation radius and $\text{Ro} = \mathcal{U}/(f_0\mathcal{L})$ is the Rossby number.

In the limit of small Ro (with $\mathcal{L} \approx \mathcal{L}_d$), these equations simplify to

$$\begin{aligned} \frac{\partial \mathbf{v}^*}{\partial t^*} + \mathbf{f}^* \times \mathbf{v}^* &= -\nabla_{\mathbf{H}}^* \eta^*, \\ \frac{\partial \eta^*}{\partial t^*} + \left(\frac{\mathcal{L}_d}{\mathcal{L}}\right)^2 \nabla_{\mathbf{H}}^* \cdot \mathbf{v}^* &= 0, \end{aligned} \quad (6.16)$$

which are simply the linearised Shallow Water equations. Transforming back to dimensional variables yields

$$\begin{aligned} \frac{\partial \mathbf{v}}{\partial t} + \mathbf{f} \times \mathbf{v} &= -g \nabla_{\mathbf{H}} \eta, \\ \frac{\partial \eta}{\partial t} + H \nabla_{\mathbf{H}} \cdot \mathbf{v} &= 0. \end{aligned} \quad (6.17)$$

The take-home message from this scaling argument is that, on the fast timescale $1/f_0$ and in the limit of small Ro with $\mathcal{L} \approx \mathcal{L}_d$, the Shallow Water equations become effectively linear and admit wave solutions.

To derive these wave solutions, it is convenient to introduce

$$\zeta = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}, \quad \delta = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = \nabla_{\mathbf{H}} \cdot \mathbf{v},$$

where ζ and δ are the vertical vorticity and horizontal divergence, respectively. In other words, we rewrite the equations in terms of (ζ, δ, η) instead of (u, v, η) .

Taking the curl and divergence of Equation 6.17 gives

$$\begin{aligned} \frac{\partial \zeta}{\partial t} + f \delta &= 0, \\ \frac{\partial \delta}{\partial t} - f \zeta &= -g \nabla_{\mathbf{H}}^2 \eta. \end{aligned} \quad (6.18)$$

Substituting $\delta = \nabla_{\mathbf{H}} \cdot \mathbf{v}$ into Equation 6.17 gives

$$\frac{\partial \eta}{\partial t} + H \delta = 0. \quad (6.19)$$

Equations Equation 6.18 and Equation 6.19 together form a closed linear system for (ζ, δ, η) , which we solve using the wave ansatz

$$\zeta = \tilde{\zeta} e^{i(k_x x + k_y y - \omega t)}, \quad \delta = \tilde{\delta} e^{i(k_x x + k_y y - \omega t)}, \quad \eta = \tilde{\eta} e^{i(k_x x + k_y y - \omega t)}.$$

Substituting into Equation 6.18–Equation 6.19 and cancelling the exponential factor leads to the matrix system

$$\begin{bmatrix} -i\omega & f & 0 \\ -f & -i\omega & -g(k_x^2 + k_y^2) \\ 0 & H & -i\omega \end{bmatrix} \begin{bmatrix} \tilde{\zeta} \\ \tilde{\delta} \\ \tilde{\eta} \end{bmatrix} = 0.$$

The dispersion relation follows from the characteristic equation

$$-i\omega(-\omega^2 + gH(k_x^2 + k_y^2)) + f(-i\omega f) = 0,$$

which gives

$$\omega = 0, \quad \omega = \pm \sqrt{f^2 + gH(k_x^2 + k_y^2)}.$$

The non-zero frequencies correspond to shallow-water waves, often referred to as *Poincaré waves*. For these waves, the phase and group velocities are

$$\mathbf{c}_p = \omega \left(\frac{k_x}{k^2}, \frac{k_y}{k^2} \right) = \frac{\omega}{k^2} \mathbf{k}, \quad \mathbf{c}_g = \left(\frac{\partial \omega}{\partial k_x}, \frac{\partial \omega}{\partial k_y} \right) = \left(\frac{gHk_x}{\omega}, \frac{gHk_y}{\omega} \right) = \frac{gH}{\omega} \mathbf{k}.$$

Thus the phase and group velocities are parallel: wave crests and wave packets move in the same direction. The ratio of group to phase speed is

$$\frac{c_g}{c_p} = \frac{gH}{c_p^2} = \frac{\mathcal{L}_d^2(k_x^2 + k_y^2)}{1 + \mathcal{L}_d^2(k_x^2 + k_y^2)},$$

which shows that wave packets move more slowly than the crests. For larger wavenumbers (shorter wavelengths), the difference between these speeds decreases.

6.3 Inertia-gravity waves in the Boussinesq equations

We now consider the wave dynamics in the context of the Boussinesq system. This leads to an important class of waves in the atmosphere and ocean: *inertia-gravity waves*. These waves are associated with two restoring effects introduced earlier: buoyancy and the Coriolis force. They can be generated in many ways, e.g. atmospheric flows impinging on topography, tides in the ocean, or wind forcing at the sea surface. Despite their diverse generation mechanisms, the waves that propagate in the ocean and atmosphere interior follow similar physical principles.

We can follow a similar scaling argument as in Section 6.2 to justify linearising the Boussinesq equations for the fast-timescale motion. We can also consider small disturbances propagating in a fluid initially at rest. If the disturbances are small, we can neglect the quadratic nonlinear terms, leading to the linearised equations. In the hydrostatic Boussinesq approximation, the linear equations are

$$\begin{aligned} \frac{\partial \mathbf{v}}{\partial t} + \mathbf{f} \times \mathbf{v} &= -\frac{1}{\rho_0} \nabla_H p, \\ b &= \frac{1}{\rho_0} \frac{\partial p}{\partial z}, \\ \nabla_H \cdot \mathbf{v} + \frac{\partial w}{\partial z} &= 0, \\ \frac{\partial b}{\partial t} + N^2 w &= 0, \end{aligned} \tag{6.20}$$

where b is buoyancy, N the buoyancy frequency, and w the vertical velocity.

Using a plane-wave ansatz for all variables,

$$\begin{aligned} u &= \tilde{u} e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}, & v &= \tilde{v} e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}, & w &= \tilde{w} e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}, \\ b &= \tilde{b} e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}, & p &= \tilde{p} e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}, \end{aligned}$$

and substituting into Equation 6.20, then cancelling the common factor $e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$, we obtain

$$\begin{aligned} -i\omega \tilde{u} - f \tilde{v} &= -\frac{1}{\rho_0} i k_x \tilde{p}, \\ -i\omega \tilde{v} + f \tilde{u} &= -\frac{1}{\rho_0} i k_y \tilde{p}, \\ \tilde{b} &= \frac{1}{\rho_0} i k_z \tilde{p}, \\ i k_x \tilde{u} + i k_y \tilde{v} + i k_z \tilde{w} &= 0, \\ -i\omega \tilde{b} + N^2 \tilde{w} &= 0. \end{aligned} \tag{6.21}$$

We eliminate $\tilde{p}$ from the first two equations using the third, and eliminate $\tilde{w}$ from the last using the fourth, to obtain the reduced system

$$\begin{bmatrix} -i\omega & -f & k_x/k_z \\ f & -i\omega & k_y/k_z \\ -N^2 k_x/k_z & -N^2 k_y/k_z & -i\omega \end{bmatrix} \begin{bmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{b} \end{bmatrix} = 0.$$

The dispersion relation follows from the characteristic equation of this matrix:

$$i\omega \left(-\omega^2 + f^2 + N^2 \frac{k_x^2 + k_y^2}{k_z^2} \right) = 0,$$

so

$$\omega = 0, \quad \omega = \pm \sqrt{f^2 + N^2 \frac{k_h^2}{k_z^2}}, \quad (6.22)$$

where $k_h = \sqrt{k_x^2 + k_y^2}$. We focus on the non-zero frequencies, which describe propagating waves.

The dispersion relation Equation 6.22 reveals key features of inertia-gravity waves. The frequency for a given wavevector $\mathbf{k}$ depends only on the ratio k_h/k_z . As illustrated schematically in Figure Figure 6.4, this ratio is equal to $\tan \theta$, where θ is the angle between the wavevector and the vertical, so

$$\frac{k_h}{k_z} = \tan(\theta).$$

Thus, all wavevectors with the same θ have the same frequency. Surfaces of constant frequency in wavenumber space are therefore double cones (Figure Figure 6.4).

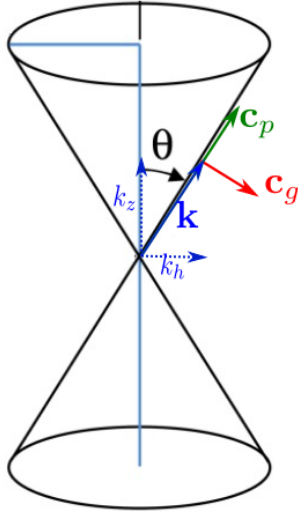


Figure 6.4: Cone of constant frequency for inertia-gravity waves with their group and phase velocity.

From the definition of group velocity Equation 6.12, we know that $\mathbf{c}_g$ is perpendicular to surfaces of constant ω , so it is normal to these cones. The phase velocity, on the other hand, is parallel to $\mathbf{k}$ and thus lies along the cone. As a result, the group and phase velocities are perpendicular to one another: energy and information propagate in a direction perpendicular to the propagation of individual crests.

We can see this explicitly by differentiating Equation 6.22 with respect to $\mathbf{k}$:

$$\mathbf{c}_g = \left(\frac{N^2 k_x / k_z^2}{\omega}, \frac{N^2 k_y / k_z^2}{\omega}, -\frac{N^2 k_h^2 / k_z^3}{\omega} \right). \quad (6.23)$$

It is straightforward to check that the inner product of this vector with $\mathbf{k} = (k_x, k_y, k_z)$ is zero, confirming that $\mathbf{c}_g \perp \mathbf{k}$.

We can repeat the analysis for the non-hydrostatic Boussinesq equations (see the earlier Boussinesq chapter) to obtain the dispersion relation

$$\omega = \pm \sqrt{f^2 \frac{k_z^2}{k^2} + N^2 \frac{k_h^2}{k^2}} = \pm \sqrt{f^2 \cos^2(\theta) + N^2 \sin^2(\theta)}, \quad (6.24)$$

where again θ is the angle from the vertical.

The hydrostatic and non-hydrostatic dispersion relations share a key feature: both depend only on θ , so the constant-frequency surfaces are the same cones, and $\mathbf{c}_p$ and $\mathbf{c}_g$ remain perpendicular. The non-hydrostatic dispersion relation Equation 6.24, however, shows an additional property: the wave frequency is bounded between f and N (for typical geophysical conditions with $N > f$). This is consistent with observations in the atmosphere and ocean.

6.4 Rossby waves

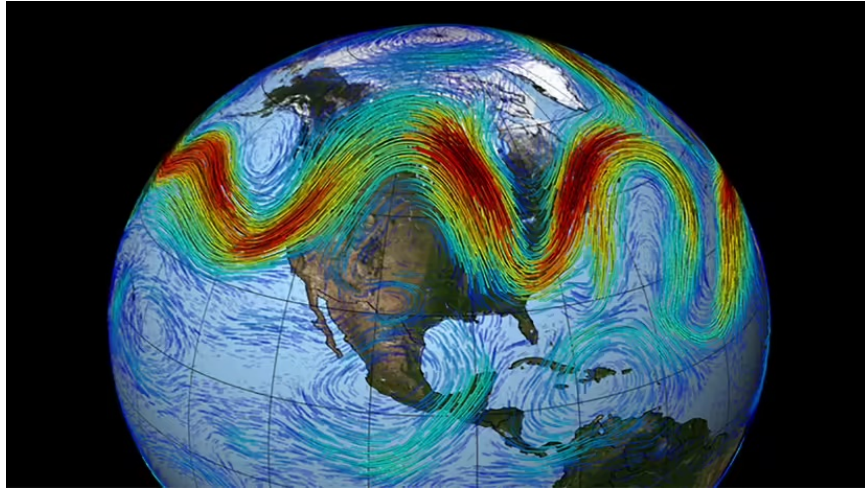


Figure 6.5: Rossby waves interacting with the jet stream. Figure adapted from an animation by NASA’s Goddard Space Flight Center.

Rossby waves are another important class of waves in the ocean and atmosphere. They have relatively large spatial scales—comparable to the planetary radius—and are therefore also called *planetary waves*. Figure 6.5 illustrates the large scale of atmospheric Rossby waves. These waves play a crucial role in shaping weather patterns, as they can transport substantial masses of air or water and give rise to cyclones and anticyclones.

The restoring mechanism for Rossby waves is the variation of the Coriolis parameter with latitude, the so-called β -effect introduced in the earlier chapter on the tangent-plane approximation. As usual, we will start from a set of linear equations and derive a dispersion relation. For Rossby waves we begin from the quasigeostrophic (QG) potential vorticity equation

$$\frac{Dq}{Dt} = 0, \quad (6.25)$$

where q is the potential vorticity (PV). We did not start from the full Boussinesq equations, because at large scales the QG approximation is appropriate. As discussed in previous chapters, PV conservation

holds for both the shallow-water and Boussinesq systems, but the expressions for q differ. Starting from Equation 6.25, one can derive Rossby waves in both systems; here we restrict attention to the shallow-water case for simplicity, noting that many properties are shared.

Recall that the QG PV for the shallow-water system is

$$q = \nabla_H^2 \psi + f_0 + \beta y - \frac{1}{\mathcal{L}_d^2} \psi, \quad (6.26)$$

where $\mathcal{L}_d = \sqrt{gH}/f_0$ is the deformation radius.

We linearise Equation 6.25 about a time-independent basic state. Unlike the inertia-gravity wave case, the basic state now includes a non-zero flow. We assume a uniform zonal (east-west) basic flow, appropriate for the strong zonal jets often observed in large-scale atmospheric and oceanic flows. We decompose the flow into a zonal mean (denoted by an overbar) and a perturbation (denoted by a prime):

$$\begin{aligned} \psi &= \bar{\psi}(y) + \psi'(x, y, t) = -\bar{u}y + \psi'(x, y, t), \\ u &= \bar{u} - \frac{\partial \psi'}{\partial y}, \quad v = \frac{\partial \psi'}{\partial x}, \end{aligned} \quad (6.27)$$

where $\bar{u}$ is a constant zonal velocity and we have used $\bar{v} = 0$. The PV Equation 6.26 becomes

$$q = \nabla_H^2 \psi' + f_0 + \beta y + \frac{\bar{u}y}{\mathcal{L}_d^2} - \frac{\psi'}{\mathcal{L}_d^2}. \quad (6.28)$$

Substituting Equation 6.27 and Equation 6.28 into Equation 6.25 and simplifying yields

$$\frac{\partial}{\partial t} \left(\nabla_H^2 \psi' - \frac{\psi'}{\mathcal{L}_d^2} \right) + \left(\bar{u} - \frac{\partial \psi'}{\partial y} \right) \frac{\partial}{\partial x} \left(\nabla_H^2 \psi' - \frac{\psi'}{\mathcal{L}_d^2} \right) + \frac{\partial \psi'}{\partial x} \frac{\partial}{\partial y} \left(\nabla_H^2 \psi' + \beta y + \frac{\bar{u}y}{\mathcal{L}_d^2} - \frac{\psi'}{\mathcal{L}_d^2} \right) = 0.$$

Neglecting products of perturbation quantities (as usual in linearisation) gives

$$\left(\frac{\partial}{\partial t} + \bar{u} \frac{\partial}{\partial x} \right) \left(\nabla_H^2 \psi' - \frac{\psi'}{\mathcal{L}_d^2} \right) + \frac{\partial \psi'}{\partial x} \left(\beta + \frac{\bar{u}}{\mathcal{L}_d^2} \right) = 0.$$

Substituting the wave ansatz $\psi' = \tilde{\psi} e^{i(k_x x + k_y y - \omega t)}$ in the above equation, we derive the dispersion relation

$$\omega = \bar{u}k_x - k_x \frac{\beta + \bar{u}/\mathcal{L}_d^2}{k_x^2 + k_y^2 + 1/\mathcal{L}_d^2}.$$

It is instructive to consider the special case of a very large deformation radius, $\mathcal{L}_d \rightarrow \infty$, which is a good approximation when the horizontal scale of the motion is much smaller than $\mathcal{L}_d$. In this limit, the dispersion relation reduces to

$$\omega = \bar{u}k_x - \frac{\beta k_x}{k_x^2 + k_y^2}.$$

The (scalar) phase speed in the x -direction is then

$$c_{p,x} = \frac{\omega}{k_x} = \bar{u} - \frac{\beta}{k_x^2 + k_y^2}. \quad (6.29)$$

i In the absence of a mean flow ($\bar{u} = 0$), the phase speed in the x -direction is negative for all wavenumbers, meaning that Rossby waves propagate westward. If we choose an eastward background flow with $\bar{u} = \beta/(k_x^2 + k_y^2)$, the phase speed becomes zero and the Rossby wave pattern is stationary.

The group velocity of Rossby waves in the limit $\mathcal{L}_d \rightarrow \infty$ is

$$\mathbf{c}_g = \left(\frac{\partial \omega}{\partial k_x}, \frac{\partial \omega}{\partial k_y} \right) = \left(\bar{u} + \frac{\beta(k_x^2 - k_y^2)}{(k_x^2 + k_y^2)^2}, \frac{2k_x k_y \beta}{(k_x^2 + k_y^2)^2} \right). \quad (6.30)$$

Unlike inertia-gravity waves, the phase and group velocities for Rossby waves are not perpendicular; instead, the angle between them depends on the wavenumber and the background flow.

Comparing Equation 6.29 and Equation 6.30, we find that the x -component of the group velocity can be written as the x -component of the phase velocity plus a positive correction:

$$\mathbf{c}_g \cdot \mathbf{e}_x = \mathbf{c}_p \cdot \mathbf{e}_x + \frac{2k_x^2 \beta}{(k_x^2 + k_y^2)^2}.$$

This is an interesting result: the group velocity of the Rossby wave is shifted eastward relative to its phase velocity. For example, consider a stationary Rossby wave (with $\mathcal{L}_d \rightarrow \infty$ and $\omega = 0$). Although the wave pattern is fixed in space, the corresponding wave packets transport energy and mass eastward.

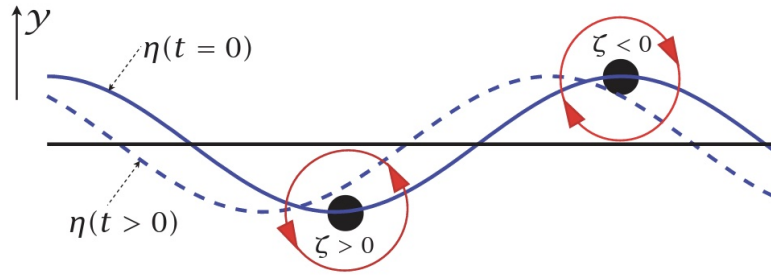


Figure 6.6: Mechanism of a Rossby wave with infinite deformation radius. An initial disturbance displaces a material line at constant latitude (black straight line) into the solid curve $\eta(t = 0)$. Conservation of absolute vorticity ($\zeta + f_0 + \beta y$) leads to the production of relative vorticity ζ for parcels moved northward or southward. The induced velocity field (arrows) then advects the fluid parcels further, and the material line evolves into the dashed curve $\eta(t > 0)$, with the phase propagating westward.

We can gain more physical intuition for Rossby waves (with large deformation radius) using a heuristic argument. Consider a line of fluid parcels at a fixed latitude y , represented by the black straight line in Figure @fig:Mechanism_RossbyWave. Suppose this material line is disturbed so that it acquires the shape $\eta(t = 0)$ (solid blue curve).

The PV in the limit $\mathcal{L}_d \rightarrow \infty$ becomes $q = \zeta + f_0 + \beta y$ and is conserved according to Equation 6.25. A parcel displaced northward (to larger y) must therefore acquire negative relative vorticity ($\zeta < 0$) to keep its absolute vorticity unchanged; a parcel displaced southward must acquire positive vorticity. As indicated in Figure @fig:Mechanism_RossbyWave, these vorticity anomalies induce a velocity field that pushes the neighbouring parcels westward, causing the disturbed material line (solid curve) to evolve into the dashed curve $\eta(t > 0)$. Thus the disturbance propagates westward, consistent with the sign of the x -component of the phase velocity in Equation 6.29 when $\bar{u} = 0$.

7 Turbulence in geophysical flows

7.1 Introduction

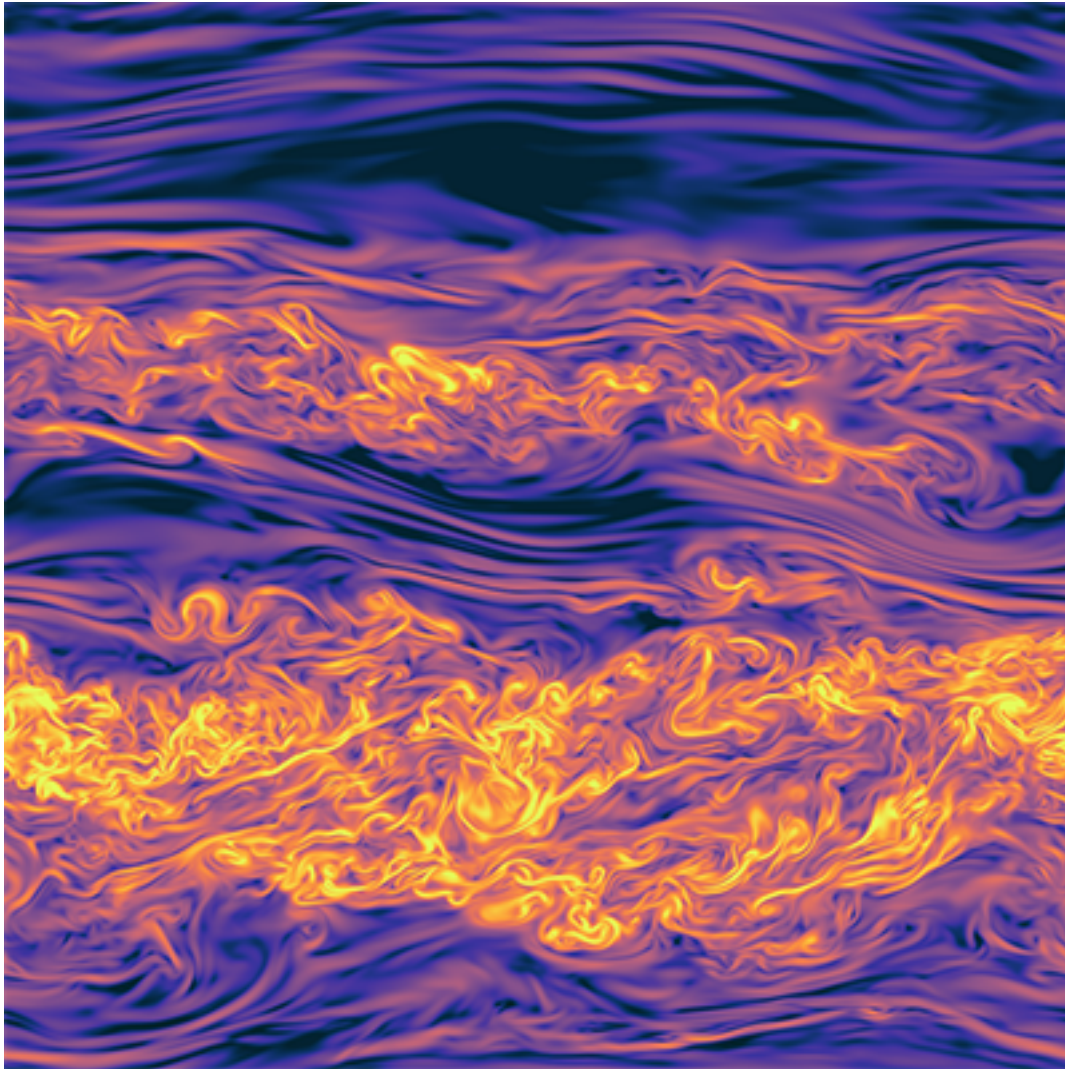


Figure 7.1: Mixing in the flow by turbulence.

I am an old man now, and when I die and go to heaven, there are two matters on which I hope for enlightenment. One is quantum electrodynamics and the other is the turbulent motion of fluids. About the former, I am really rather optimistic.

There is a physical problem that is common to many fields, that is very old, and that has not been solved. It is not the problem of finding new fundamental particles, but something left over from a long time ago—over a hundred years. Nobody in physics has really been able to

analyze it mathematically satisfactorily, in spite of its importance to the sister sciences. It is the analysis of circulating or turbulent fluids.

The latter quotation is attributed to Richard Feynman,¹ one of the most celebrated physicists of the twentieth century, while the former originates from Sir Horace Lamb nearly a century earlier. These and similar remarks frequently appear in opening paragraphs of talks, articles, and books on turbulence—not as a celebration of progress, but as an acknowledgement of the persistent challenges this subject poses. Beyond their wit, they reflect the inherent difficulty of turbulence in physics, a difficulty that begins with the challenge of defining it.

Before attempting such a definition, let us recall an important dimensionless quantity that is central to the study of turbulence: the **Reynolds number**

$$\text{Re} = \frac{\mathcal{U}\mathcal{L}}{\nu}, \quad (7.1)$$

where $\mathcal{U}$ is a characteristic velocity scale, $\mathcal{L}$ a characteristic length scale, and ν the kinematic viscosity. The inverse of $\mathcal{U}/\nu$ has units of length and represents a scale at which molecular viscosity becomes significant—one of the smallest dynamically relevant length scales in the flow. By contrast, the scale $\mathcal{L}$ typically denotes the largest dynamically active scales, such as the forcing scale, domain size, or the scale of the most energetic vortices. The Reynolds number therefore quantifies the ratio between the largest and smallest active spatial scales. Atmospheric and oceanic flows exhibit enormously large Reynolds numbers (often of order 10^{11}), reflecting the breadth of scales they involve and the complexity of their dynamics.

As leading turbulence textbooks acknowledge, any precise definition of turbulence is at best difficult and arguably impossible. Following their approach, we characterise turbulence through its principal features:

Randomness and chaos: Turbulent motion is highly irregular and unpredictable. A hallmark of chaotic systems is the sensitivity to initial conditions: arbitrarily small perturbations may evolve into drastically different states. Consequently, deterministic prediction of particle trajectories or instantaneous field values (e.g. velocity or pressure) is infeasible. Instead, turbulence is studied statistically. Remarkably, despite the apparent randomness, turbulent flows exhibit universal statistical properties; several of these are explored below.

Enhanced mixing and diffusivity: Turbulence greatly increases the transport of momentum, heat, and scalar quantities. One may imagine turbulence as an “idealised whisk” stirring velocity, temperature, and other fields, thereby enhancing fluxes and accelerating mixing.

A wide range of spatial scales: Turbulence excites a continuum of Fourier modes. A flow with only a small number of active modes is not turbulent, even if chaotic. Because Re is (in a loose sense) the ratio of the outer to inner scales, large Reynolds numbers are associated with a wide spectral range. In this sense, “turbulence” is inherently a large-Reynolds-number phenomenon.

Nonlinear interactions: Without nonlinear terms, Fourier modes evolve independently. The Navier–Stokes equations, however, contain quadratic nonlinearities. Turbulence requires these nonlinear interactions to be dynamically significant, allowing energy and other invariants to be transferred between scales.

In the atmosphere and ocean, the largest scales exceed the dissipation scale by many orders of magnitude. Between them lies a vast range of intermediate scales that interact nonlinearly. This richness underpins both the complexity of weather and the limits of weather predictability. Thus, it is fair to state that both the atmosphere and ocean are turbulent systems, although the form and regime of turbulence vary across spatial and temporal scales.

¹From “The Feynman Lectures on Physics”, Vol. 1; .

Before describing specific turbulence regimes in geophysical flows, we first review a central mathematical tool: the *Fourier transform*.

7.2 Fourier transform

Because turbulence involves nonlinear energy exchange between scales, the Fourier transform plays a central role. The Fourier transform of a function $f(\mathbf{x})$, denoted $\hat{f}$ or $\mathcal{F}[f]$, is defined as

$$\hat{f}(\mathbf{k}) = \mathcal{F}[f(\mathbf{x})](\mathbf{k}) = \int_{-\infty}^{\infty} f(\mathbf{x}) e^{-i\mathbf{k}\cdot\mathbf{x}} d\mathbf{x}, \quad (7.2)$$

where $\mathbf{k} = (k_x, k_y, k_z)$ is the wavevector. The inverse transform is

$$f(\mathbf{x}) = \mathcal{F}^{-1}[\hat{f}(\mathbf{k})](\mathbf{x}) = \frac{1}{(2\pi)^3} \int_{-\infty}^{\infty} \hat{f}(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{x}} d\mathbf{k}.$$

We frequently use two key properties:

Linearity:

$$\mathcal{F}[af + bg] = a\mathcal{F}[f] + b\mathcal{F}[g].$$

Differentiation:

$$\mathcal{F}\left[\frac{\partial f}{\partial x}\right] = ik_x \hat{f}, \quad \mathcal{F}\left[\frac{\partial f}{\partial y}\right] = ik_y \hat{f}, \quad \mathcal{F}\left[\frac{\partial f}{\partial z}\right] = ik_z \hat{f}.$$

These properties make Fourier space a natural setting in which to analyse turbulence and energy transfers across scales.

7.3 Theory of 3D isotropic turbulence

The modern theory of isotropic turbulence centres on the transfer of energy from large to small vortices—a process called the *energy cascade*. Richardson encapsulated this idea in his well-known verse:

*Big whirls have little whirls that feed on their velocity,
and little whirls have lesser whirls and so on to viscosity.*

Large eddies transfer energy to somewhat smaller eddies, which transfer it to still smaller eddies, and so on, until the energy reaches sufficiently small scales that it is dissipated by viscosity. This is illustrated schematically in the left panel of Figure 7.2.

Building on this idea, Kolmogorov introduced a phenomenological theory predicting how energy is distributed across scales. His theory—often considered the most significant single contribution to turbulence—applies directly to small-scale atmospheric and oceanic motions and forms the basis for several geophysical turbulence theories.

Kolmogorov’s theory relies on four principal assumptions: (i) *isotropy*, (ii) *homogeneity*, (iii) *large Reynolds number*, and (iv) *locality of energy transfer*. Isotropy means that the statistical properties of the flow are invariant under rotations of the coordinate system; homogeneity means translation invariance. These assumptions hold away from boundaries or when boundary layers are thin relative to the domain.

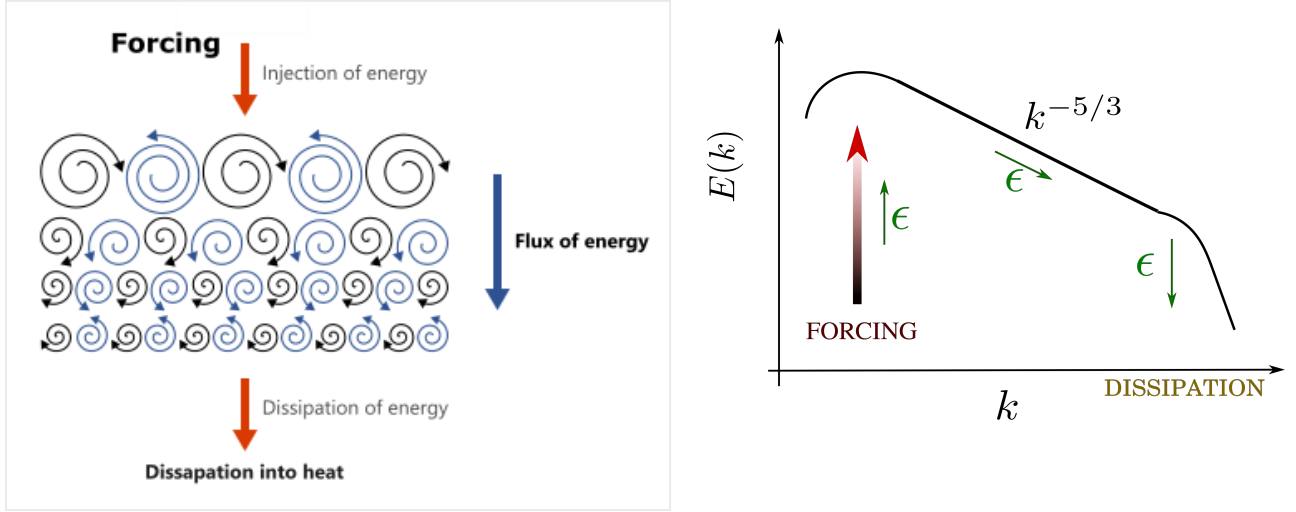


Figure 7.2: Left: schematic picture of energy cascade by vortices of different sizes. Right: the energy spectrum of isotropic 3D turbulence.

To describe the distribution of energy among Fourier modes, let $\hat{\mathbf{u}}(\mathbf{k})$ be the Fourier transform of the velocity field. The modal kinetic energy at wavenumber $\mathbf{k}$ is

$$\mathcal{E}(\mathbf{k}) = \frac{1}{2} \hat{\mathbf{u}}(\mathbf{k}) \cdot \hat{\mathbf{u}}(\mathbf{k})^*,$$

and Parseval's theorem yields

$$E_{\text{tot}} = \frac{1}{V} \int_V \frac{1}{2} \mathbf{u}(\mathbf{x}) \cdot \mathbf{u}(\mathbf{x}) d\mathbf{x} = \int \mathcal{E}(\mathbf{k}) d\mathbf{k}. \quad (7.3)$$

Isotropy implies that modes with equal $|\mathbf{k}| = k$ have statistically identical energies. This motivates the definition of the *energy spectrum*

$$E(k) = \oint_{S_k} \mathcal{E}(\mathbf{k}) d\mathbf{s}, \quad (7.4)$$

where S_k is the sphere of radius k in Fourier space. Therefore,

$$E_{\text{tot}} = \int_0^\infty E(k) dk. \quad (7.5)$$

Thus $E(k) dk$ represents the kinetic energy per unit mass contained in wavenumbers between k and $k + dk$.

If the flow is statistically stationary, energy conservation dictates that the rate at which energy is injected into the flow, ϵ , must equal the rate at which it is dissipated by viscosity. The energy spectrum may depend on several quantities, including the forcing F , the dissipation ν , the energy flux across each wavenumber ϵ , and the wavenumber k :

$$E = f(F, \epsilon, k, \nu). \quad (7.6)$$

Kolmogorov's final two assumptions now enter. Locality asserts that energy is transferred mainly between modes of comparable wavenumber. Because forcing acts at large scales and viscosity at small scales, a wide inertial subrange exists where neither forcing nor dissipation acts directly. In this range the spectrum cannot depend on F or ν , leaving

$$E = f_i(\epsilon, k). \quad (7.7)$$

Dimensional analysis then suggests the form

$$E = C \epsilon^m k^n, \quad (7.8)$$

where C is a dimensionless constant. With dimensions

$$[E] = L^3 T^{-2}, \quad [\epsilon] = L^2 T^{-3}, \quad [k] = L^{-1},$$

matching powers of L and T yields $m = 2/3$ and $n = -5/3$, giving the celebrated Kolmogorov spectrum,

$$E(k) = C \epsilon^{2/3} k^{-5/3}. \quad (7.9)$$

This $k^{-5/3}$ law describes the energy spectrum of any (sufficiently isotropic) turbulent flow, regardless of the underlying fluid or environment. Laboratory experiments and oceanic observations have verified this law. The right panel of Figure 7.2 shows the typical 3D isotropic turbulent spectrum.

7.4 2D turbulence

While neither the atmosphere nor the ocean is exactly two-dimensional, the theory of 2D turbulence forms the conceptual foundation of geostrophic turbulence, which successfully describes many large-scale geophysical statistics. This connection arises because quasi-geostrophic (QG) flows and strictly 2D flows share important invariants and exhibit qualitatively similar nonlinear behaviour.

Consider the 2D incompressible Navier–Stokes equations without rotation, viscosity, or external forcing:

$$\begin{aligned} \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} &= -\frac{1}{\rho} \nabla_H p, \\ \nabla \cdot \mathbf{v} &= 0. \end{aligned} \quad (7.10)$$

Taking the curl yields the vorticity equation

$$\frac{D\zeta}{Dt} = 0, \quad (7.11)$$

where ζ is the vertical vorticity.

A fundamentally different phenomenology emerges in 2D turbulence due to the conservation of vorticity alongside energy. If vorticity ζ is conserved for each fluid particle, the volume-averaged *enstrophy*, defined as

$$Z_{\text{tot}} = \frac{1}{V} \int_V \frac{1}{2} \zeta(\mathbf{x}) \zeta(\mathbf{x}) d\mathbf{x} = \frac{1}{2} \int \hat{\zeta}(\mathbf{k}) \hat{\zeta}(\mathbf{k})^* d\mathbf{k}, \quad (7.12)$$

is also conserved. Here, $\hat{\zeta}$ denotes the Fourier transform of ζ .

Analogous to the energy spectrum defined in Equation 7.4, the enstrophy spectrum is given by

$$Z(k) = \frac{1}{2} \int_{S_k} \hat{\zeta}(\mathbf{k}) \hat{\zeta}(\mathbf{k})^* d\mathbf{s}, \quad (7.13)$$

where S_k represents circles of constant k . The enstrophy spectrum relates to the energy spectrum via the definition of vorticity:

$$Z(k) = k^2 E(k). \quad (7.14)$$

Combining Equation 7.14 with Equation 7.13 and Equation 7.12 gives

$$Z_{\text{tot}} = \int_0^\infty Z(k) dk = \int_0^\infty k^2 E(k) dk. \quad (7.15)$$

Similar to energy, enstrophy is injected into the flow by forcing and dissipated by viscosity. When there is a separation between the forcing and dissipation scales, enstrophy can cascade across intermediate scales. By analogy with the energy flux ϵ , the enstrophy flux η is defined as the rate of enstrophy transfer across scales (which equals the rate of enstrophy injection and dissipation in stationary flows). In inertial ranges—scales unaffected directly by forcing or dissipation—both ϵ and η are expected to remain constant (i.e., neither ϵ nor η is a function of k in inertial ranges to maintain the conservation of energy and enstrophy). Using dimensional similarity, one can derive the relationship

$$\eta = \mathcal{A} k^2 \epsilon, \quad (7.16)$$

where $\mathcal{A}$ is a dimensionless constant.

If energy is conserved and the flow is statistically stationary, the energy flux ϵ remains constant across wavenumbers in the inertial range. A similar argument holds for enstrophy conservation and η . However, from Equation 7.16, if ϵ is constant, η would necessarily vary with k , unless $\mathcal{A} = 0$. If $\mathcal{A} = 0$, then $\eta = 0$ as well. Conversely, if η is constant, ϵ must vary with k . Consequently, energy and enstrophy fluxes cannot coexist in the same inertial range in 2D isotropic turbulence.

When $\epsilon = 0$ and $\eta \neq 0$, the energy spectrum depends solely on the enstrophy flux and wavenumber. Dimensional analysis similar to what was presented for Equation 7.9 yields

$$E(k) = \mathcal{C}_\eta \eta^{2/3} k^{-3}, \quad (7.17)$$

where $\mathcal{C}_\eta$ is a dimensionless constant. On the other hand, if $\epsilon \neq 0$ and $\eta = 0$, the energy spectrum follows Kolmogorov's formula Equation 7.9:

$$E(k) = \mathcal{C}_\epsilon \epsilon^{2/3} k^{-5/3}. \quad (7.18)$$

This raises an important question: which inertial range dominates in a given 2D turbulent flow? This question is closely tied to whether energy or enstrophy cascades upscale or downscale. Before presenting a mathematical argument for the direction of these cascades, we first develop physical intuition by examining a numerical experiment.

We consider a 2D Navier–Stokes simulation governed by Equation 7.10 with an initial distribution of vortices that has a thin energy spectrum localized near the wavenumber $k_i = 20$, and we set the forcing term $F = 0$. The vorticity field of the initial condition in physical space is shown in Figure 7.3 (a), with its energy spectrum plotted on a logarithmic scale in Figure 7.4.

The flow is then allowed to evolve over time. Figure 7.3 illustrates the evolution of this flow in physical space at various times. Initially, the flow's energy is concentrated at smaller vortices, consistent with the energy spectrum peaking at the intermediate (or relatively small) scale of $1/k_i$. Over time, smaller vortices transfer their energy to larger ones that gradually become the most energetic structures. This process continues, forming increasingly larger vortices (see panels (a), (b) and (c) of Figure 7.3). This behaviour contrasts sharply with 3D turbulence, where energy cascades from large to small scales. In 2D turbulence, energy moves from smaller to larger scales, as evidenced by the energy spectrum's peak shifting toward smaller wavenumbers (larger scales) over time, as shown in Figure 7.4. This energy transfer to larger scales is referred to as the *inverse cascade*, distinguishing it from the forward cascade characteristic of 3D turbulence.

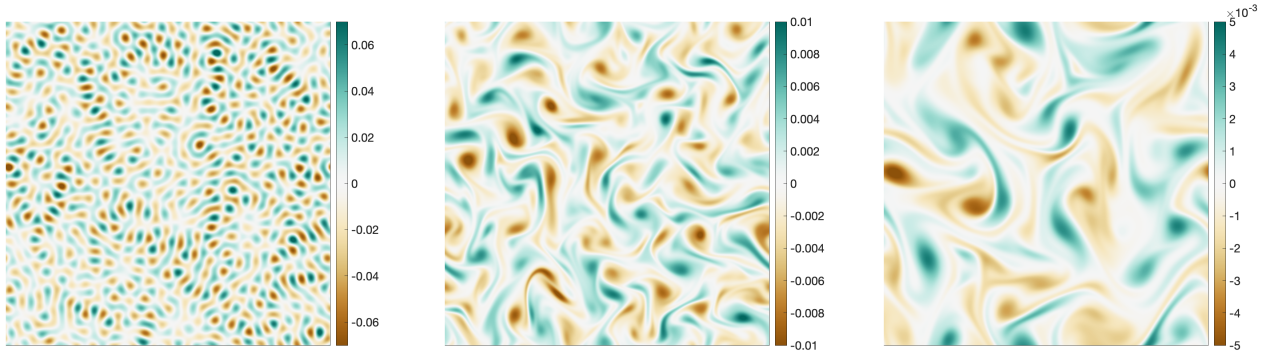


Figure 7.3: Evolution of a decaying turbulent 2D flow at initial time (a), intermediate time (b), and the end of the simulation (c). Colours represent the vorticity field.

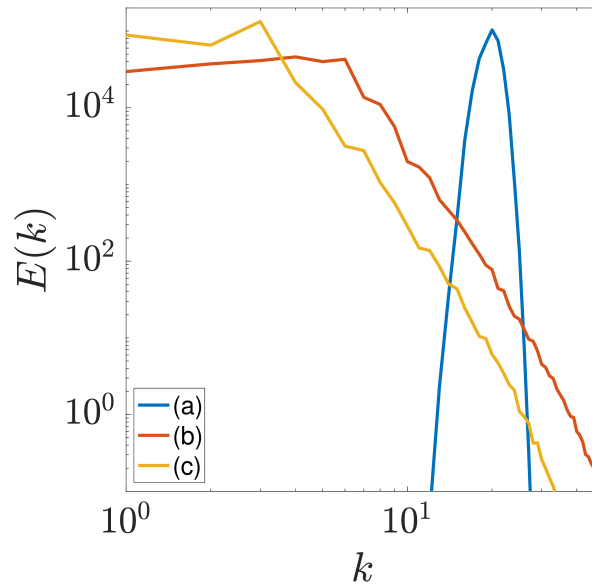


Figure 7.4: Energy spectra corresponding to the panels shown in Figure 7.3.

To put these observations on a firmer mathematical foundation, let us analyze the evolution of energy distribution in wavenumber space. If the initial energy spectrum is sufficiently narrow, we can define the characteristic wavenumber of the initial condition k_i by

$$k_i = \frac{\int k E(k) dk}{\int E(k) dk} = \frac{\int k E(k) dk}{E_{\text{tot}}}. \quad (7.19)$$

We want to see how the spectrum evolves in time. A fundamental characteristic of turbulence is mixing, which leads to the spread of this narrow band of energy in wavenumber space. In other words, due to strong nonlinear interactions in turbulence, the energy is naturally transferred to other scales. We can quantify the spreading out of the energy distribution with

$$W = \int (k - k_i)^2 E(k) dk,$$

which can be expanded to

$$\begin{aligned} W &= \int k^2 E(k) dk - 2k_i \int k E(k) dk + k_i^2 \int E(k) dk \\ \Rightarrow W &= Z_{\text{tot}} - k_i^2 E_{\text{tot}}, \end{aligned}$$

where we used Equation 7.5, Equation 7.15 and Equation 7.19 to arrive at the final expression:

$$W = Z_{\text{tot}} - k_i^2 E_{\text{tot}}. \quad (7.20)$$

Since energy and enstrophy are conserved in 2D turbulence, $dE_{\text{tot}}/dt = dZ_{\text{tot}}/dt = 0$. Then, differentiating Equation 7.20 with respect to time gives

$$\frac{dk_i^2}{dt} = -\frac{1}{E_{\text{tot}}} \frac{dW}{dt}.$$

The spreading of the initial narrow energy band implies $dW/dt > 0$, leading to $dk_i^2/dt < 0$. Thus, k_i decreases over time, indicating that the typical length scale of vortices increases, consistent with the numerical results shown in Figure 7.3.

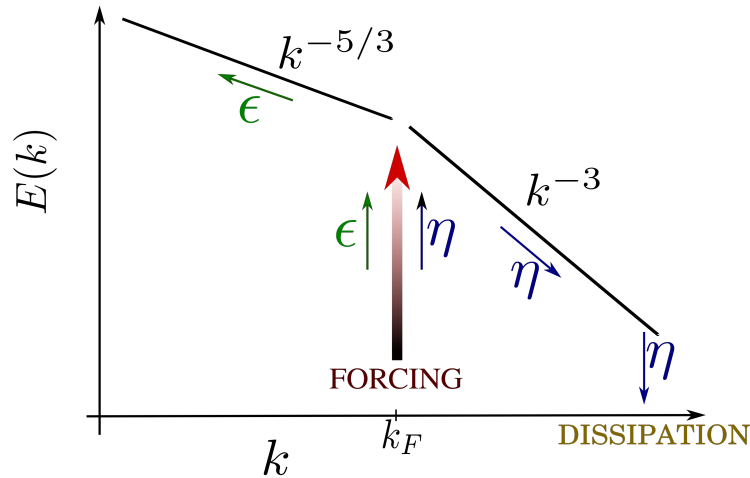


Figure 7.5: Energy spectrum of 2D turbulence with both forward cascade of enstrophy and inverse cascade of energy.

Now, let us return to the forced problem, equipped with the insight that energy in 2D turbulence is transferred to larger scales. If the flow is forced at an intermediate scale k_F , we expect energy to cascade upscale for $k < k_F$, resulting in an energy spectrum following Equation 7.18. Conversely, enstrophy, which is also conserved in 2D flows, must cascade downscale for $k > k_F$ (as its flux cannot co-exist with energy flux in the range of $k < k_F$), leading to an energy spectrum consistent with Equation 7.17. This behaviour, known as the double or dual cascade, is depicted schematically in Figure 7.5, where the cascade directions and the corresponding slopes of spectra are shown.

This picture of forced 2D turbulence is commonly referred to as the *Kraichnan–Batchelor* theory, named after the two prominent scientists who developed it. To achieve the statistical stationarity assumed in Equation 7.18, a mechanism must dissipate energy at scales much larger than the forcing scale. A plausible physical mechanism is the scale-independent dissipation characteristic of Ekman layers near boundaries as long as their effects remain negligible within the inertial range.

7.5 Geostrophic turbulence

The seminal paper by Charney (1971) established the theory of geostrophic turbulence. The motivation stemmed from a paradox: observational spectra showed a k^{-3} slope at large atmospheric scales—consistent with 2D turbulence—yet the atmosphere is clearly not two-dimensional in this regime.

The theory relies on the QG system introduced in Section~Chapter 5, which accurately describes large-scale atmospheric and oceanic flows. For simplicity, assume constant N and f . The total QG energy, consisting of horizontal kinetic plus available potential energy, is

$$E_{\text{tot}} = \frac{1}{V} \int_V \left(u^2 + v^2 + \frac{b^2}{N^2} \right) d\mathbf{x} = \frac{1}{V} \int_V \left(|\nabla\psi|^2 + \frac{f_0^2}{N^2} \psi_z^2 \right) d\mathbf{x}. \quad (7.21)$$

The QG potential vorticity,

$$q = \zeta + \frac{f_0}{N^2} \frac{\partial b}{\partial z} = \nabla_H^2 \psi + \frac{f_0^2}{N^2} \psi_{zz}, \quad (7.22)$$

is materially conserved,

$$\frac{Dq}{Dt} = \frac{\partial q}{\partial t} + \mathbf{v} \cdot \nabla_H q = 0, \quad (7.23)$$

implying conservation of *potential enstrophy*,

$$Z_{\text{tot}} = \frac{1}{V} \int_V q^2 d\mathbf{x}. \quad (7.24)$$

QG flow is neither 2D nor 3D isotropic. To make its invariants resemble those of 2D turbulence, Charney proposed a vertical coordinate stretching:

$$\tilde{z} = \frac{N}{f} z.$$

In the stretched coordinate, Equation 7.23 becomes

$$\frac{\partial}{\partial t} \nabla_*^2 \psi + u \frac{\partial}{\partial x} \nabla_*^2 \psi + v \frac{\partial}{\partial y} \nabla_*^2 \psi = 0, \quad (7.25)$$

where

$$\nabla_* = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial \tilde{z}} \right)$$

is an isotropic gradient operator. The energy and potential enstrophy become

$$E_{\text{geo}} = \int |\nabla_* \psi|^2 d\mathbf{x}, \quad (7.26)$$

and

$$Z_{\text{geo}} = \int (\nabla_*^2 \psi)^2 d\mathbf{x}. \quad (7.27)$$

These have the same mathematical form as the 2D energy and enstrophy invariants, except that ∇_H has been replaced by ∇_* and the integrals are now three-dimensional.

Therefore, the same arguments that apply to 2D turbulence apply to QG turbulence in the stretched coordinates. In particular, potential enstrophy cascades forward, and the associated energy spectrum in the “synoptic” range is

$$E(\tilde{k}) = C_{\text{geo}} \eta^{2/3} \tilde{k}^{-3}, \quad (7.28)$$

where $\tilde{k} = (k_x^2 + k_y^2 + (f/N)^2 k_z^2)^{1/2}$.

Assuming isotropy in ∇_* space, the kinetic and potential energy components,

$$E_x, E_y, E_z,$$

each inherit the same k^{-3} slope. Transforming back to physical coordinates implies that both horizontal kinetic and available potential energy spectra in the atmosphere follow this -3 scaling.

7.6 Atmospheric energy spectrum

We now illustrate how turbulence theories explain the observed atmospheric energy spectrum. Figure 7.6, taken from Nastrom and Gage 1985, is a landmark observational result covering a broad range of scales, obtained from over 6000 commercial aircraft flights in the Global Atmospheric Sampling Program (GASP). Later studies using improved datasets reproduced similar spectra.

In the synoptic range (corresponding to wavelengths up to roughly 500 km), the zonal velocity, meridional velocity, and potential temperature all exhibit a k^{-3} slope. This is precisely the scaling predicted by geostrophic turbulence theory in the forward-cascading enstrophy subrange (Section~Section 7.5). Because E_x , E_y , and E_z all share the same k^{-3} slope, the consistency across all three observed spectra supports the applicability of QG dynamics at synoptic scales.

At wavelengths smaller than 500 km, the spectrum transitions sharply to a shallower $k^{-5/3}$ slope (sometimes interpreted as -2). While the k^{-3} regime is well-understood, the physical explanation of the $k^{-5/3}$ range remains an active area of research.

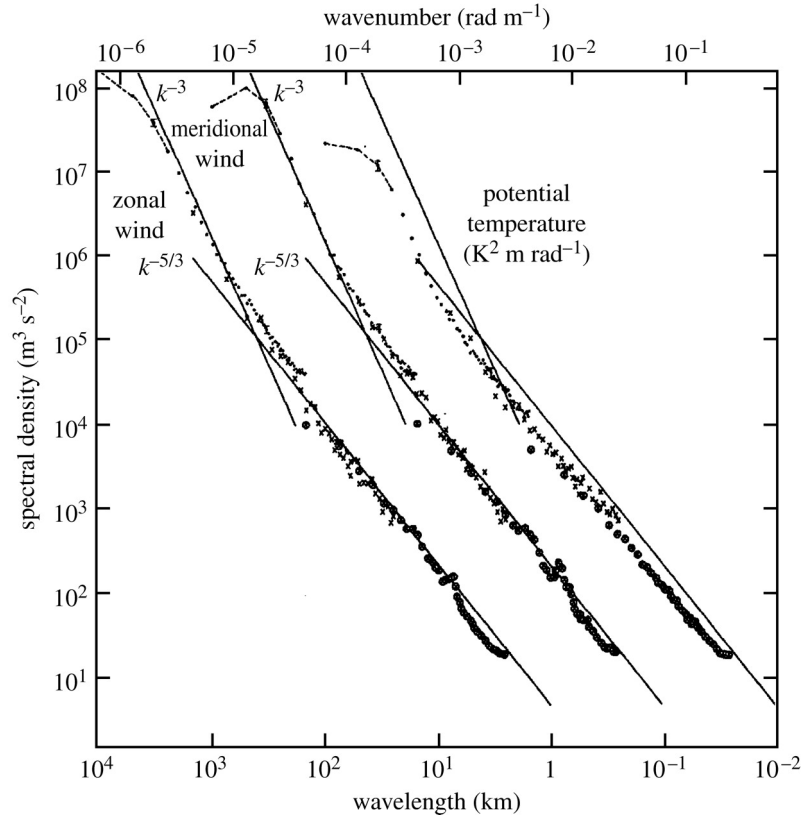


Figure 7.6: Wavenumber spectra of zonal (u) and meridional (v) velocities and potential temperature. Dashed lines indicate slopes of -3 and $-5/3$. For visual clarity, the v and θ spectra are horizontally shifted by one and two decades, respectively.